

Probability Theory

Lecture Notes

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Probability Theory

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Course Probability Theory

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Probability Theory

Lecture 01

Introduction to Probability

Date: Wednesday, 05 August 2026

Instructor: Dr. Nishant Chandgotia

1.1 Motivation and Some Problems

Question 1. Sleeping Beauty Problem

A princess is put into a deep sleep by a powerful drug administered by a villain. The drug ensures that she will remain asleep for two years, except for possible brief awakenings determined by the outcome of a fair coin toss.

The villain tosses a fair coin.

- If the coin lands **Heads**, the princess is awakened *once*, briefly, at the end of the **first year**. After this awakening, her memory of having awakened is completely erased, and she is put back to sleep until the end of the second year.
- If the coin lands **Tails**, the princess is not awakened after the first year. Instead, she is awakened *once*, briefly, at the end of the **second year**.

Whenever the princess awakens, she has no information about the current date or the passage of time. Furthermore, if she had awakened previously, all memory of that awakening has been erased. Consequently, upon waking, she cannot distinguish between the two possible awakening scenarios.

Suppose that the princess wakes up and is immediately asked the following question:

“What probability should you assign to the event that the coin landed Heads?”

What should the princess answer?

Solution:

1. $\frac{1}{2}$ No information exchanged so probability should be $\frac{1}{2}$, If repeated independently a million times, half chances each.
2. $\frac{2}{3}$ There are possibilities of this. Proportion of waking during Heads is $\frac{2}{3}$, and $\dots \frac{1}{3}$
3. $\frac{3}{4}$

Villain tosses once, H $\rightarrow$ Villain tosses again, Two Possibilities H and T, so,

- HH $\rightarrow \frac{1}{4}$
- HT $\rightarrow \frac{1}{4}$

- $T \rightarrow \frac{1}{2}$

Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

Question 2. Monty Hall Problem

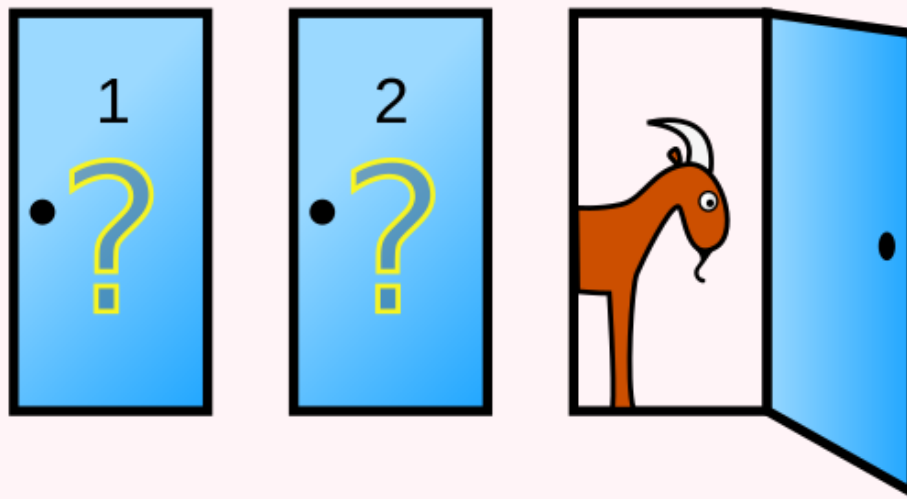


Figure 1: Monty Hall Problem

There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is $P(\text{chosen door}) = \frac{1}{3}$, while the probability that the car is behind one of the other two doors is $P(\text{other two doors}) = \frac{2}{3}$.

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore, $P(\text{Win by staying}) = \frac{1}{3}$, $P(\text{Win by switching}) = \frac{2}{3}$. Hence, switching gives twice the probability of winning compared to staying.

Remark

The Above question (Monty Hall Problem) is not ill posed.

Summary

- Sleeping Beauty Problem
- Monty Hall Problem

End of Lecture

Probability Theory

Lecture 02

Introduction to Probability

Date: Friday, 07 August 2026

Instructor: Dr. Nishant Chandgotia

2.1 Some Introductory Idea

Suppose we toss a fair coin n -times. $HTH HTTH \dots T$

What is the Probability of the above sequence ? Its- $\sim \frac{1}{2^n}$

If we have all sequences with k -heads then $P(k\text{-heads}) = \frac{\binom{n}{k}}{2^n}$, and it will be maximum when $k = \frac{n}{2}$.

If $x_1, x_2, \dots, x_k$ are randomly independently chosen, then $\sum_{i=1}^k x_i \sim \frac{r+1}{2} \cdot k$, if they are chosen from the uniform distribution on $\{1, 2, \dots, r\}$.

Now Stirling's Formula, $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, for very large n .

Now,

$$\frac{P(\text{number of heads} = k)}{P(\text{number of heads} = k+1)} = \frac{k+1}{n-k}.$$

So, $\frac{k+1}{n-k} > 1 \iff k > \frac{n-1}{2}$, and $\frac{k+1}{n-k} < 1 \iff k < \frac{n-1}{2}$.

As $\binom{n}{k}$ is maximum when $k = \frac{n}{2}$, therefore, $\frac{\binom{n}{n/2}}{2^n} \sim \frac{2}{\sqrt{2\pi n}}$ (By using Stirling's Formula). It

would be unjustified to say $\# \text{Heads} = \frac{n}{2}$ with high probability.

Suppose $k = \alpha n$, $\alpha \neq \frac{1}{2}$.

Then $\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(-\alpha \log \alpha - (1-\alpha) \log(1-\alpha) - \log 2)}$.

Let, $H(\alpha) = -\alpha \log \alpha - (1-\alpha) \log(1-\alpha)$, be Boltzman Shanon Entropy (Mathematical theory of Communications).

Hence,

$$\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(H(\alpha) - H(\frac{1}{2}))}.$$

Now, $H(\alpha) \leq H(\frac{1}{2})$, by using Jensen's inequality as $\log$ is a concave function.

Equivalently, $-\alpha \log \alpha - (1-\alpha) \log(1-\alpha) < \ln 2$.

Therefore, $\frac{\binom{n}{k}}{2^n} \rightarrow 0$ exponentially.

Now,

$$P(\#\text{Heads} \in ((\alpha - \varepsilon)n, (\alpha + \varepsilon)n)) = \sum_{k=(\alpha-\varepsilon)n}^{(\alpha+\varepsilon)n} \frac{\binom{n}{k}}{2^n}, \quad \text{where } \varepsilon > 0, \quad \frac{1}{2} \notin (\alpha - \varepsilon, \alpha + \varepsilon).$$

Also,

$$\frac{2n\varepsilon}{2^n} \binom{n}{(\alpha - \varepsilon)n} \leq \cdots \leq \frac{2n\varepsilon}{2^n} \binom{n}{(\alpha + \varepsilon)n}.$$

Therefore, the asymptotic bounds are $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n(H(\alpha-\varepsilon)-H(\frac{1}{2}))}$ and $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n(H(\alpha+\varepsilon)-H(\frac{1}{2}))}$.

Therefore, $P(\#\text{Heads} \in ((\alpha - \varepsilon)n, (\alpha + \varepsilon)n)) \rightarrow 0$ exponentially as $n \rightarrow \infty$.

Hence,

$$P\left(\#\text{Heads} \in n\left(\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon\right)\right) \rightarrow 1.$$

This is called the **Large Deviation Phenomenon**.

$P(\text{staying close to mean}) \rightarrow 1$,

whereas $P(\text{deviating from the mean significantly}) \rightarrow 0$ exponentially.

In our case, the rate function is $H(\alpha) - H(\frac{1}{2})$.

Let $f_n = \frac{\#\text{Heads}}{n}$. Then $f_n \rightarrow f$ in measure.

Therefore,

$$P\left(\left|\frac{\#\text{Heads}}{n} - \frac{1}{2}\right| > \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (1)$$

We call this **convergence in probability**.

Later we shall use the **Borel–Cantelli Lemma** to prove from (1) that $\frac{\#\text{Heads}}{n} \rightarrow \frac{1}{2}$ almost surely.

The sample space is $\Omega = \{H, T\}^{\mathbb{N}}$. We equip Ω with the product topology, where each copy of $\{H, T\}$ carries the discrete topology. The corresponding Borel σ -algebra, $\mathcal{F} = \mathcal{B}(\Omega)$, is generated by the cylinder sets. The basis of the product topology consists of the cylinder sets, namely sets of the form $\{\omega \in \Omega : \omega_{i_1} = a_1, \dots, \omega_{i_k} = a_k\}$, where $i_1, \dots, i_k$ are finitely many coordinates and each $a_j \in \{H, T\}$.

If we only specify the first n coordinates, they look like $\{\omega : \omega_1 = a_1, \dots, \omega_n = a_n\}$, which are the basic cylinders.

The complete probability space is $(\Omega, \mathcal{F}, \mu) = (\{H, T\}^{\mathbb{N}}, \mathcal{B}(\{H, T\}^{\mathbb{N}}), \mu)$,

Recall,

$$P\left(\#\text{Heads} = \frac{n}{2}\right) \sim \frac{2}{\sqrt{2\pi n}}.$$

Suppose

$$c > 0,$$

and let $|k - \frac{n}{2}| < c\sqrt{n}$.

Then

$$P(\text{\#Hheads} = k) = \frac{n!}{(n-k)!k!} \frac{1}{2^n}.$$

Using Stirling's formula,

$$\begin{aligned} P(\text{\#Hheads} = k) &\sim \frac{1}{\sqrt{\pi n/2}} \cdot \frac{1}{\left(2 - \frac{2k}{n}\right)^{n-k} \left(\frac{2k}{n}\right)^k} \\ &= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \cdot \frac{1}{\left(1 + \left(1 - \frac{2k}{n}\right)\right)^{n-k} \left(1 + \left(\frac{2k}{n} - 1\right)\right)^k}. \end{aligned}$$

Hence,

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(n-2k)^2}{n}\right).$$

Equivalently, $= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(k - \frac{n}{2})^2}{2 \cdot \frac{n}{4}}\right)$, which resembles the normal distribution.

$$\mu = \frac{1}{2}, \quad \sigma = \frac{1}{2}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi\sigma^2n}} \exp\left(-\frac{(k - \mu n)^2}{2\sigma^2n}\right).$$

We can prove, $P\left(a < \frac{\text{\#Hheads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x) dx$, as $n \rightarrow \infty$. (**Central Limit Theorem / De Moivre–Laplace**)

End of Lecture

2.2 Quiz, 10 August 2026

Quiz 1

1. Let $p < \frac{1}{2}$. Suppose we toss n coins independently such that the probability of heads is p and tails is $q = 1 - p$.

For what value of k is the probability of getting k heads maximised?

2. For probability vectors $P = (p_1, p_2, \dots, p_n)$, $Q = (q_1, q_2, \dots, q_n)$, where $p_i, q_i > 0$, $\sum_i p_i = \sum_i q_i = 1$, define the Kullback–Leibler divergence by

$$D_{KL}(P\|Q) = \sum_i p_i \log \left(\frac{p_i}{q_i} \right).$$

Use Jensen's inequality to show that $D_{KL}(P\|Q) \geq 0$ with equality if and only if $P = Q$.

3. Let $\alpha \in (0, 1)$. Prove that the number of heads equalling $n\alpha$ is asymptotically

$$\frac{1}{\sqrt{2\pi n\alpha(1-\alpha)}} \exp(-nD_{KL}(P\|Q)),$$

where $P = (\alpha, 1 - \alpha)$, $Q = (p, q)$.

Probability Theory

Lecture 03

Introduction to Probability

Date: Wednesday, 12 August 2026

Instructor: Dr. Nishant Chandgotia

3.1 Central Limit Theorem and Convergence

Suppose we toss a fair coin independently n times. The Central Limit Theorem suggests that

$$P\left(a < \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x) dx, \quad n \rightarrow \infty,$$

where, for a fair coin, and

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

3.1.1 Galton Board

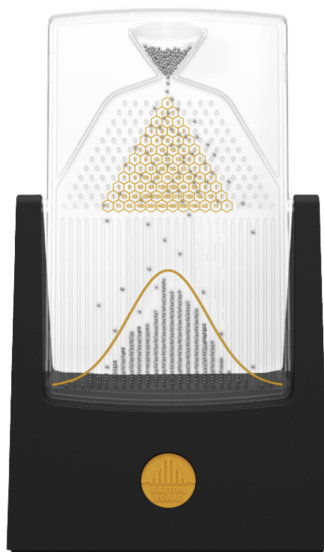


Figure 2: Galton board. Source: [Wikimedia Commons](#).

The Galton board gives a visual representation of a process whose distribution approaches a normal distribution.

3.1.2 Random Walk

A *random walk* is a stochastic process in which the position changes randomly at each step.

For example, consider a person starting at the origin. At every step, the person tosses a fair coin:

$$\begin{cases} H & \longrightarrow \text{move one step to the right,} \\ T & \longrightarrow \text{move one step to the left.} \end{cases}$$

This gives the classical one-dimensional symmetric random walk.

There are several natural questions:

Questions

- Will the person return to the origin?
- Will the person escape to infinity?
- How often will the person return to the origin?

3.1.3 Convergence in Distribution

We know,

$$\frac{\#\text{Heads}}{n} \longrightarrow \frac{1}{2}.$$

Now define

$$f_n = \frac{\#\text{Heads} - \mu n}{\sigma \sqrt{n}}.$$

Then the Central Limit Theorem states that

$$P(a < f_n < b) \longrightarrow \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

This leads to the notion of *weak convergence*, or *convergence in distribution*.

A natural question is whether convergence in distribution implies almost-sure convergence.

Important

Convergence in distribution does not, in general, imply almost-sure convergence.

3.1.4 An Example

Consider the following random process.

Start with 1.

Toss a fair coin independently at every step:

$$\begin{cases} H & \longrightarrow \text{multiply by } 2, \\ T & \longrightarrow \text{multiply by } -2. \end{cases}$$

After n steps, define

$$f_n = \frac{\text{product after } n \text{ steps}}{2^n}.$$

Since every multiplication contributes either 2 or -2 ,

$$f_n \in \{-1, 1\}.$$

Moreover,

$$P(f_n = 1) = \frac{1}{2}, \quad P(f_n = -1) = \frac{1}{2}.$$

Thus the distribution of f_n is the same for every n .

However, f_n does not converge almost surely.

Questions

- Does f_n converge almost surely?
- Does f_n converge in L^1 ?
- Does f_n converge in probability?

Remark

The example illustrates that convergence in distribution is a weaker notion of convergence than almost-sure convergence.

3.2 Objects in Probability

There is a close analogy between measure theory and probability theory.

Table 1: Analogy between Measure Theory and Probability

Measure Theory	Probability
Measure space: $(\Omega, \mathcal{B}, \mu)$	Probability space: $(\Omega, \mathcal{B}, \mathbb{P})$
σ -algebra $\mathcal{B}$: $A \in \mathcal{B}$, where $\mu(A)$ is defined.	Event: $A \in \mathcal{B}$, for which $\mathbb{P}(A)$ is defined.
Elements: $\omega \in \Omega$	Outcomes / samples: $\omega \in \Omega$
Measurable function:	Random variable:
$f : (\Omega, \mathcal{B}) \longrightarrow (S, \rho)$	$X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$
L^1 -integrability:	Mean / expectation:
$\int_{\Omega} f d\mu < \infty$	$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}$
L^2 -norm:	Variance:
$\ f\ _{L^2}^2 = \int_{\Omega} f ^2 d\mu$	$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
Push-forward measure:	Distribution of X :
$f_*\mu(A) = \mu(f^{-1}(A))$	$\nu(A) = \mathbb{P}(X^{-1}(A))$
Fourier transform:	Characteristic function:
$\hat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$	$\Phi_X(t) = \mathbb{E}[e^{itX}]$
Convergence in measure	Convergence in probability
Almost everywhere (a.e.)	Almost surely (a.s.)
Convergence in L^1	Convergence in mean.

3.3 Sample Space and Sigma Algebra

The first object in probability is the *sample space*.

Ω = the universe of all possible outcomes.

For example, for a single coin toss, $\Omega = \{H, T\}$.

For a bus arriving at a non-negative time, $\Omega = [0, \infty)$.

For infinitely many coin tosses, $\Omega = \{H, T\}^{\mathbb{N}}$.

An element of this space is an infinite sequence $\omega = (\omega_1, \omega_2, \dots)$, $\omega_i \in \{H, T\}$.

3.3.1 Sigma Algebra

Let $\mathcal{B}$ denote the σ -algebra of events.

For a single coin toss, $\mathcal{B} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.

For the infinite product space $\{H, T\}^{\mathbb{N}}$, the relevant σ -algebra is generated by cylinder sets.

3.3.2 Cylinder Sets

Suppose we are interested only in the first finitely many tosses.

Let $F \subset \mathbb{N}$ be a finite set and let $a : F \rightarrow \{H, T\}$ be a fixed assignment of heads and tails.

Define $[a] = \{\omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F\}$.

The set $[a]$ specifies the outcomes of the coordinates belonging to the finite set F , while leaving all other coordinates unrestricted.

Such sets are called *cylinder sets*.

They form a basis for the product topology on $\{H, T\}^{\mathbb{N}}$.

We want to be able to assign a probability $\mathbb{P}([a])$ to every cylinder set.

For a fair coin, if $|F| = k$, then $\mathbb{P}([a]) = \frac{1}{2^k}$.

End of Lecture

Probability Theory

Lecture 04

Introduction to Probability

Date: Friday, 14 August 2026

Instructor: Dr. Nishant Chandgotia

4.1 Sample Spaces, Random Variables and Their Distributions

4.1.1 Sample Space

The first object in probability is the *sample space* Ω , which is the collection of all possible outcomes.

1. **Simple coin toss** $\Omega = \{H, T\}$.
2. **Arrival time of a bus** Depending on the context, we may take $\Omega = [0, 1]$ or $\Omega = [0, \infty)$.
3. **Infinite sequence of coin tosses** $\Omega = \{H, T\}^{\mathbb{N}}$.

4.1.2 σ -Algebra

1. A σ -algebra $\mathcal{B}$ is the collection of events whose probabilities we can measure.
For a simple coin toss, $\mathcal{B} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.
2. For a subset of $\mathbb{R}$, we are naturally led to the *Borel σ -algebra*.
3. For the infinite product space $\{H, T\}^{\mathbb{N}}$, consider a finite set $F \subset \mathbb{N}$ and a function $a : F \rightarrow \{H, T\}$.

Define the corresponding cylinder set by

$$[a] = \left\{ \omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F \right\}.$$

Cylinder sets form the basic building blocks for the product σ -algebra.

4.1.3 Probability Measure

A probability measure is a measure $\mathbb{P} : \mathcal{B} \rightarrow [0, 1]$ such that $\mathbb{P}(\Omega) = 1$.

1. For a fair coin, $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$.

2. For the uniform probability measure on $[0, 1]$,

$$\mathbb{P}([a, b]) = b - a, \quad 0 \leq a \leq b \leq 1.$$

3. For independent coin tosses, if $[a]$ is a cylinder set determined by $|F|$ coordinates, then

$$\mathbb{P}([a]) = 2^{-|F|}.$$

4. More generally, for a coin with $\mathbb{P}(H) = p$, $\mathbb{P}(T) = 1 - p$, we have

$$\mathbb{P}([a]) = p^{\#\text{Heads}}(1 - p)^{\#\text{Tails}}.$$

Question: Why does this define a probability measure on $\{H, T\}^{\mathbb{N}}$?

4.1.4 Non-measurable Sets

Important

Not every subset of $\{H, T\}^{\mathbb{N}}$ is measurable.

The construction of non-measurable sets is related to the existence of suitable equivalence relations and translation-invariant measures.

A useful analogy is the classical construction for $\mathbb{R}$.

4.1.4.1 Proof Idea on $\mathbb{R}$

The general strategy is:

1. Find a countable subgroup of $\mathbb{R}$.
2. Find a Borel measure satisfying translation invariance:

$$\mu(A + x) = \mu(A), \quad \forall x \in \mathbb{R}, \quad A \in \mathcal{B}.$$

4.1.4.2 Analogue for the Infinite Coin-Toss Space

Consider $\left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}}$.

One can consider the set

$$G = \left\{ x \in \left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}} : x(i) = 0 \text{ for all sufficiently large } i \right\}.$$

The idea is to check that G is a countable subgroup and then investigate translation invariance of the probability measure on cylinder sets.

In particular, for a cylinder set $[a]$,

$$\mathbb{P}([a]) = 2^{-|F|}.$$

4.2 Random Variables

A random variable is a measurable function from the probability space to another measurable space.

For example, for infinitely many coin tosses, $f_n : \{H, T\}^{\mathbb{N}} \rightarrow [0, 1]$, defined by

$$f_n(\omega) = \frac{1}{n} \sum_{i=1}^n X_i(\omega),$$

where

$$X_i(\omega) = \begin{cases} 1, & \omega_i = H, \\ 0, & \omega_i = T. \end{cases}$$

We would like to understand the limit

$$f(\omega) = \lim_{n \rightarrow \infty} f_n(\omega).$$

4.2.1 A Random Variable Detecting the Existence of a Limit

Question 1. Random Variable for Existence of Limit

Given a probability space and a sequence of random variables (f_n) , define a random variable which captures whether or not $\lim_{n \rightarrow \infty} f_n(\omega)$ exists.

For a real-valued sequence, the limit exists precisely when $\limsup_{n \rightarrow \infty} f_n(\omega) = \liminf_{n \rightarrow \infty} f_n(\omega)$ and both quantities are finite.

Define

$$\tilde{f}(\omega) = \begin{cases} \limsup_{n \rightarrow \infty} f_n(\omega) - \liminf_{n \rightarrow \infty} f_n(\omega), & \text{if both are finite,} \\ +\infty, & \text{otherwise.} \end{cases}$$

Then define

$$f'(\omega) = \begin{cases} 1, & \tilde{f}(\omega) = 0, \\ 0, & \tilde{f}(\omega) \neq 0. \end{cases}$$

Thus

$$f'(\omega) = \mathbf{1}_{\{\lim_{n \rightarrow \infty} f_n(\omega) \text{ exists as a finite real number}\}}.$$

Remark

The precise formulation and notation may differ from the presentation in Durrett.

4.3 Distribution of a Random Variable

Let $X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$ be a random variable.

The distribution of X is the push-forward of $\mathbb{P}$ under X : $X_*\mathbb{P}$.

More explicitly, the distribution μ_X of X is the measure on (S, ρ) defined by

$$\mu_X(A) = \mathbb{P}\left(X^{-1}(A)\right), \quad A \in \rho.$$

It is also common to write

$$\mathbb{P}(X \in A) := \mathbb{P}\left(X^{-1}(A)\right).$$

Consider the random variables X_1 and X_2 arising from the independent coin-toss example above.

Then X_1 and X_2 have the same distribution: $X_1 \stackrel{d}{=} X_2$.

However, they are not the same random variable.

Indeed, $X_1(\omega) \neq X_2(\omega)$ for some ω .

Also, (X_1, X_2) is itself a random variable, taking values in the appropriate product space.

Remark

Having the same distribution does not imply that two random variables are equal.
In symbols,

$$X = Y \implies X \stackrel{d}{=} Y,$$

but in general,

$$X \stackrel{d}{=} Y \not\implies X = Y.$$

4.4 Some Useful Definitions:

Let $(X_n)_{n \geq 1}$ and X be random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

4.4.1 Convergence in Probability

Definition 1. Convergence in Probability

We say that X_n **converges in probability** to X , denoted by

$$X_n \xrightarrow{\mathbb{P}} X,$$

if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0.$$

4.4.2 Convergence in Distribution

Definition 2. Convergence in Distribution

We say that X_n **converges in distribution** (or *in law*) to X , denoted by

$$X_n \xrightarrow{d} X,$$

if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for every continuity point x of the distribution function F_X of X .

Here,

$$F_{X_n}(x) = \mathbb{P}(X_n \leq x), \quad F_X(x) = \mathbb{P}(X \leq x).$$

4.4.3 Convergence in L^1

Definition 3. L_1 Convergence

We say that X_n **converges to X in L^1** , denoted by

$$X_n \xrightarrow{L^1} X,$$

if

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|] = 0.$$

Equivalently,

$$\|X_n - X\|_1 = \mathbb{E}[|X_n - X|] \longrightarrow 0.$$

4.4.4 Almost Sure Convergence

Definition 4. Almost Sure Convergence

We say that X_n **converges almost surely** to X , denoted by

$$X_n \xrightarrow{\text{a.s.}} X,$$

if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

Equivalently,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1.$$

That is, the sequence $X_n(\omega)$ converges to $X(\omega)$ for almost every $\omega \in \Omega$.

4.5 Types of Random Variables

Random variables can be classified according to their distributions.

1. **Discrete random variables**
 2. **Continuous random variables**
 - (a) Absolutely continuous random variables.
 - (b) Singular random variables.
- These classifications are determined by the distribution of the random variable.
 - A discrete random variable has an *atomic distribution*.

4.5.1 Examples of Discrete Random Variables

1. **Bernoulli random variable** Let $p \in [0, 1]$. We write $X \sim \text{Ber}(p)$ if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

This notation means that X has the Bernoulli distribution with parameter p .

2. **Binomial random variable**

Let $n \in \mathbb{N}$, $p \in [0, 1]$. We write $X \sim \text{Bin}(n, p)$

if

$$\mathbb{P}(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, \dots, n.$$

3. **Poisson random variable**

Let $\lambda > 0$. We write $X \sim \text{Poi}(\lambda)$

if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

4. **Geometric random variable**

Let $p \in (0, 1)$. Under the convention that X takes values in $\{1, 2, \dots\}$, we write $X \sim \text{Geo}(p)$

if

$$\mathbb{P}(X = k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

End of Lecture

4.6 Quiz, 17 August 2026

Quiz 2

1. Convergence in probability implies convergence almost surely.
2. Convergence in distribution implies convergence in probability.
3. Convergence almost surely implies convergence in distribution.
4. Let $X, Y, Z \sim \text{Ber}(1/2)$ such that $X \perp Y$, $Y \perp Z$ and $Z \perp X$. Then

$$\mathbb{P}(X = 1, Y = 1, Z = 1) = \frac{1}{8}.$$

Probability Theory

Lecture 05

Introduction to Probability

Date: Wednesday, 19 August 2026

Instructor: Dr. Nishant Chandgotia

5.1 Continuous Random Variables

Question 1. Quiz-Question-Discussion

The last question from the quiz showed that pairwise independence and mutual independence are different notions. What is the distinction between the two?

5.1.1 Continuous Random Variables

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (S, \rho)$ be a random variable.

In many cases, the state space is one of $\mathbb{R}$, $\mathbb{R}^d$, equipped with the Borel σ -algebra.

A random variable is called *continuous* when its distribution does not have atoms. In particular, for a real-valued continuous random variable, $\mathbb{P}(X = s) = 0$, $\forall s \in \mathbb{R}$.

The distribution of X is the push-forward of $\mathbb{P}$ via X :

5.1.2 Absolutely Continuous and Singular Distributions

Let $\mu_X = X_*(\mathbb{P})$ denote the distribution of X .

Definition 1. Absolutely Continuous

The distribution μ_X is said to be *absolutely continuous* with respect to Lebesgue measure λ if there exists a non-negative function $f \in L^1(\mathbb{R})$ such that $\int_{\mathbb{R}} f d\lambda = 1$ and $\mu_X(A) = \mathbb{P}(X \in A) = \int_A f(x) d\lambda(x)$ for every Borel set $A \subseteq \mathbb{R}$.

The function f is called the *density* of X .

Remark

The statement that the distribution of a random variable is absolutely continuous means that its distribution is absolutely continuous with respect to Lebesgue measure.

On the other hand, a distribution is called *singular* with respect to Lebesgue measure if it is concentrated on a set of Lebesgue measure zero.

More precisely, if $\nu = X_*(\mathbb{P})$, then ν is singular with respect to Lebesgue measure if there exists a Borel set $A \subseteq \mathbb{R}$ such that

$$\nu(A) = 1, \quad \lambda(A) = 0.$$

5.1.3 Examples of Continuous Random Variables

1. Uniform random variable

The uniform distribution on $[0, 1]$ has density

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

2. Normal random variable

Let

$$X \sim N(\mu, \sigma^2), \quad \sigma > 0.$$

Its density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Hence, for $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

3. Exponential random variable

Let

$$\lambda > 0, \quad X \sim \text{Exp}(\lambda).$$

The random variable takes values in $[0, \infty)$ and has density

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Thus, for $0 \leq a < b$,

$$\mathbb{P}(a < X < b) = \int_a^b \lambda e^{-\lambda x} dx.$$

The exponential distribution has the *memoryless property*:

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

5.1.4 A Singular Continuous Distribution

Consider $\Omega = \{0, 1\}^{\mathbb{N}}$ with the product σ -algebra and the usual product probability measure.

Define

$$X(\omega) = \sum_{n=1}^{\infty} \frac{\omega_n}{2^n}.$$

Then

$$X : \{0, 1\}^{\mathbb{N}} \longrightarrow [0, 1].$$

Question 2.

What is the distribution of X ?

For a fixed sequence $\tilde{\omega} : \{1, \dots, n\} \longrightarrow \{0, 1\}$, consider the corresponding dyadic interval

$$I_{\tilde{\omega}} = \left[\sum_{i=1}^n \frac{\tilde{\omega}_i}{2^i}, \sum_{i=1}^n \frac{\tilde{\omega}_i}{2^i} + \frac{1}{2^n} \right].$$

The event that the first n coordinates agree with $\tilde{\omega}$ is the cylinder set $[\tilde{\omega}]$, and

$$\mathbb{P}(X \in I_{\tilde{\omega}}) = \mathbb{P}([\tilde{\omega}]) = 2^{-n}.$$

Since the length of the dyadic interval is also

$$|I_{\tilde{\omega}}| = 2^{-n},$$

we obtain, for dyadic intervals $I \subseteq [0, 1]$,

$$\mathbb{P}(X \in I) = |I|.$$

This leads to the uniform distribution on $[0, 1]$.

5.1.5 The Cantor Distribution

Now consider

$$\tilde{X}(\omega) = \sum_{n=1}^{\infty} \frac{2\omega_n}{3^n}, \quad \omega_n \in \{0, 1\}.$$

This is the ternary expansion obtained by using only the digits 0 and 2.

Consequently,

$$\tilde{X}(\omega) \in C$$

almost surely, where C denotes the middle-thirds Cantor set.

Recall that

$$\lambda(C) = 0.$$

Thus the distribution of $\tilde{X}$ is concentrated on a set of Lebesgue measure zero.

Important

The Cantor distribution is a singular continuous distribution. It is continuous in the sense that

$$\mathbb{P}(\tilde{X} = \theta) = 0$$

for every $\theta \in \mathbb{R}$, but its distribution is singular with respect to Lebesgue measure because it is concentrated on the Cantor set.

To see why there are no atoms, consider the ternary expansion

$$\theta = \sum_{n=1}^{\infty} \frac{\theta_n}{3^n}.$$

If some $\theta_i \notin \{0, 2\}$, then

$$\mathbb{P}(\tilde{X} = \theta) = 0.$$

If $\theta_i \in \{0, 2\}$ for every i , then fixing the first n digits determines a cylinder set of probability 2^{-n} .

Therefore,

$$\mathbb{P}(\tilde{X} = \theta) \leq 2^{-n} \text{ for every } n, \text{ and hence } \mathbb{P}(\tilde{X} = \theta) = 0.$$

5.1.6 Bernoulli Convolution

Consider $\lambda \in \left(\frac{1}{2}, 1\right)$ and let $\omega = (\omega_1, \omega_2, \dots) \in \{-1, 1\}^{\mathbb{N}}$.

Define

$$X_{\lambda}(\omega) = \sum_{n=1}^{\infty} \omega_n \lambda^n.$$

This is an example of a *Bernoulli convolution*.

Question 3.

For which values of λ is the distribution of X_{λ} absolutely continuous?

This question is related to the classical problems surrounding Bernoulli convolutions and the Erdős conjecture.

Question 4.

What distribution should we use to model a given random phenomenon?

5.2 Bertrand's Paradox

Question 5.

Suppose we have a unit disc and an inscribed equilateral triangle. Consider a uniformly chosen chord of the disc.

What is the probability that the chord has length greater than the side length of the triangle?

Let AB denote the side of the inscribed equilateral triangle and CD denote the randomly chosen chord. We want to determine

$$\mathbb{P}(CD > AB).$$

The difficulty is that the phrase *uniformly chosen chord* does not specify a unique probability model.

Remark

Bertrand's paradox illustrates that the answer depends on how "uniformly" is interpreted.

5.2.1 Method 1: Choosing Two Endpoints

Choose the two endpoints C and D independently and uniformly from the circumference of the circle.

Under this model,

$$\mathbb{P}(CD > AB) = \frac{1}{3}.$$

5.2.2 Method 2: Choosing the Midpoint

Choose a radius uniformly and then choose a point M uniformly along that radius. Construct the unique chord whose midpoint is M .

Under this model,

$$\mathbb{P}(CD > AB) = \frac{1}{2}.$$

The difference in the two answers demonstrates the ambiguity in the phrase "uniformly chosen chord."

Question 6.

What is the distribution of the chord length under each of these probability models?

5.3 Distribution as a Push-Forward

The distribution of a random variable is simply the push-forward of the underlying probability measure.

Suppose

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is a real-valued random variable.

Its distribution is

$$\nu = X_*(\mathbb{P}).$$

Therefore,

$$\mathbb{P}(X \in [a, b]) = \nu([a, b]) = \mathbb{P}\left(X^{-1}([a, b])\right).$$

5.3.1 Change of Variables

Suppose

$$\Omega = [a, b]$$

with Lebesgue probability measure, and suppose X is a sufficiently regular monotone function.

Then the distribution of X can be obtained from the push-forward measure.

If X is invertible, then

$$\mathbb{P}(X \in [c, d]) = \mathbb{P}\left(X^{-1}([c, d])\right).$$

For a monotone differentiable function,

$$X^{-1}(d) - X^{-1}(c) = \int_c^d (X^{-1})'(t) dt.$$

Using

$$(X^{-1})'(t) = \frac{1}{X'(X^{-1}(t))},$$

we obtain

$$\mathbb{P}(X \in [c, d]) = \int_c^d \frac{1}{X'(X^{-1}(t))} dt,$$

whenever the required regularity and monotonicity assumptions hold.

5.3.2 Example: Chord Length

Method 1.

Let

$$\Omega = [0, 1]$$

and define

$$X(\theta) = |e^{i\pi\theta} - 1|.$$

Since

$$|e^{i\alpha} - 1| = 2 \left| \sin \frac{\alpha}{2} \right|,$$

we have

$$X(\theta) = 2 \sin \left(\frac{\pi\theta}{2} \right), \quad 0 \leq \theta \leq 1.$$

Thus

$$X : [0, 1] \longrightarrow [0, 2].$$

Its inverse is

$$X^{-1}(x) = \frac{2}{\pi} \arcsin \left(\frac{x}{2} \right).$$

Therefore,

$$\frac{d}{dx} X^{-1}(x) = \frac{1}{\pi \sqrt{1 - \frac{x^2}{4}}}.$$

Hence the density of X is

$$f_X(x) = \frac{1}{\pi \sqrt{1 - \frac{x^2}{4}}}, \quad 0 < x < 2.$$

Method 2.

Alternatively, consider

$$X : [0, 1] \longrightarrow [0, 2]$$

defined by

$$X(\omega) = 2\sqrt{1 - \omega^2}.$$

Here,

$$X^{-1}(x) = \sqrt{1 - \frac{x^2}{4}}, \quad 0 \leq x \leq 2.$$

The corresponding density is obtained by differentiating the inverse appropriately.

Important

The central point of Bertrand's paradox is that specifying the word "uniformly" is not enough. A probability model must specify the underlying sample space and probability measure.

Different choices of the probability space and probability measure can produce different distributions for the same geometric quantity.

End of Lecture

Probability Theory

Lecture 06

Introduction to Probability

Date: Friday, 21 August 2026

Instructor: Dr. Nishant Chandgotia

6.1 Distribution Functions and Random Variables

Random variables are measurable maps

$$X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

We define the cumulative distribution function (C.D.F.) of X by

Definition 1. Cumulative Distribution Function

The *cumulative distribution function* of a random variable X is the function

$$F_X : \mathbb{R} \longrightarrow [0, 1]$$

defined by

$$F_X(t) = \mathbb{P}(X \leq t).$$

Remark

Naming. This is what Durrett calls the *distribution function*.

The C.D.F. completely determines the distribution of a real-valued random variable.

If $X \sim \mu$, then the C.D.F. of X is given by

$$F_X(t) = \mu((-\infty, t]).$$

Conversely, suppose that a C.D.F. F_X is given. We would like to construct the corresponding probability measure μ_X on $\mathbb{R}$.

For $a < b$, define

$$\mu_X((a, b]) = F_X(b) - F_X(a).$$

One can check that this extends to a probability measure on $\mathbb{R}$.

6.1.1 Properties of the C.D.F.

Let F_X be the C.D.F. of a random variable X .

1. F_X is an increasing function.
2. $\lim_{t \rightarrow \infty} F_X(t) = 1$, $\lim_{t \rightarrow -\infty} F_X(t) = 0$.
3. F_X is right-continuous:

$$\lim_{t \downarrow s} F_X(t) = F_X(s).$$

Indeed,

$$\lim_{t \downarrow s} F_X(t) = \lim_{t \downarrow s} \mathbb{P}(X \leq t) = \mathbb{P}(X \leq s),$$

by continuity from above.

Remark

The C.D.F. need not be left-continuous.

In fact,

$$\mathbb{P}(X = t) > 0$$

if and only if

$$F_X(t) - F_X(t^-) > 0,$$

where

$$F_X(t^-) = \lim_{s \uparrow t} F_X(s).$$

Moreover,

$$F_X(t) - F_X(t^-) = \mathbb{P}(X = t).$$

The remaining standard properties are:

4. F_X has at most countably many discontinuities.
5. F_X is Borel measurable.
6. $F_X(t^-) = \mathbb{P}(X < t)$.

6.2 The Probability Measure Associated with a C.D.F.

We now check the construction

$$\mu_X((a, b]) = F_X(b) - F_X(a)$$

and see why it defines a probability measure.

The natural collection of sets to consider first is the semiring

$$\mathcal{S} = \{(a, b] : a < b\}.$$

Suppose

$$(a, b] \subseteq \bigcup_{i=1}^{\infty} (a_i, b_i].$$

We want to verify the appropriate countable subadditivity/additivity property.

For each interval,

$$\mu_X((a_i, b_i]) = F_X(b_i) - F_X(a_i).$$

Hence

$$\sum_{i=1}^{\infty} \mu_X((a_i, b_i]) = \sum_{i=1}^{\infty} (F_X(b_i) - F_X(a_i)).$$

The key point is the right-continuity and monotonicity of F_X , which allow the interval $(a, b]$ to be approximated by finitely many of the covering intervals.

6.2.1 Finite Additivity

Suppose

$$(a, b] = \bigcup_{i=1}^n (a_i, b_i]$$

where the intervals are disjoint and ordered so that

$$a = a_1 < b_1 = a_2 < b_2 = a_3 < \cdots < a_n < b_n = b.$$

Then

$$\begin{aligned} \mu_X((a, b]) &= F_X(b) - F_X(a) \\ &= \sum_{i=1}^n (F_X(b_i) - F_X(a_i)) \\ &= \sum_{i=1}^n \mu_X((a_i, b_i]). \end{aligned}$$

Thus μ_X is finitely additive on the semiring $\mathcal{S}$.

6.2.2 Countable Additivity

The same idea extends to countable disjoint unions.

Suppose

$$(a, b] = \bigcup_{i=1}^{\infty} (a_i, b_i],$$

where the intervals are pairwise disjoint.

For the finite partial unions,

$$\bigcup_{i=1}^n (a_i, b_i] \subseteq (a, b],$$

and finite additivity gives

$$\sum_{i=1}^n \mu_X((a_i, b_i]) \leq \mu_X((a, b]).$$

Passing to the limit,

$$\sum_{i=1}^{\infty} \mu_X((a_i, b_i]) \leq \mu_X((a, b]).$$

The reverse inequality follows from the covering property, together with the continuity properties of F_X .

Consequently,

$$\mu_X((a, b]) = \sum_{i=1}^{\infty} \mu_X((a_i, b_i]).$$

Hence μ_X is countably additive on the semiring.

6.3 Correspondence Between C.D.F.s and Probability Measures

Let

$$\mathcal{F} = \left\{ F : \begin{array}{l} F \text{ satisfies the properties} \\ \text{of a C.D.F.} \end{array} \right\}$$

and let

$$\mathcal{P} = \{\text{probability measures on } \mathbb{R}\}.$$

The correspondence $F \mapsto \mu_F$ is bijective.

In other words, a probability measure on $\mathbb{R}$ determines a unique C.D.F., and a C.D.F. determines a unique probability measure.

6.4 Constructing a Random Variable from a C.D.F.

Question 1. Constructing a Random Variable from a C.D.F.

Suppose we are given a function F satisfying the properties of a C.D.F. Is there a random variable X such that

$$F = F_X?$$

We want to construct such a random variable explicitly.

The idea is to use the probability space

$$([0, 1], \mathcal{B}([0, 1]), \lambda),$$

where λ denotes Lebesgue measure.

6.4.1 Case 1: F is Continuous and Strictly Increasing

Suppose first that

$$F : (\mathbb{R}) \longrightarrow (0, 1)$$

is bijective and has no discontinuities.

Claim. We can take

$$(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), \lambda).$$

For any probability measure μ , we seek an algorithmic way of constructing

$$X : \Omega \longrightarrow \mathbb{R}$$

such that

$$X \sim \mu.$$

Define

$$X = F^{-1}.$$

Then, for every $t \in \mathbb{R}$,

$$\begin{aligned} \mathbb{P}(X \leq t) &= \mathbb{P}(F^{-1} \leq t) \\ &= \mathbb{P}\left(\left\{\omega : F^{-1}(\omega) \leq t\right\}\right) \\ &= \mathbb{P}(\{\omega : \omega \leq F(t)\}) \\ &= \mathbb{P}((0, F(t)]) \\ &= F(t). \end{aligned}$$

Therefore,

$$F_X(t) = F(t),$$

and hence

$$F_X = F.$$

6.4.2 Case 2: F Has a Jump

Case 2.

Suppose F has a jump at t .

Let

$$a = F(t^-), \quad b = F(t).$$

Define

$$X(\omega) = \begin{cases} F^{-1}(\omega), & \omega < a, \\ t, & a \leq \omega \leq b, \\ F^{-1}(\omega), & \omega > b. \end{cases}$$

Thus the entire interval $(a, b]$ is mapped to the single point t .

This creates the required atom:

$$\mathbb{P}(X = t) = b - a = F(t) - F(t^-).$$

Todo: Add the corresponding graph.

6.4.3 Case 3: F Has a Flat Portion

Case 3.

Suppose F is constant on an interval $[a, b]$.

Let

$$\theta = F(a) = F(b).$$

Define

$$X(\omega) = \begin{cases} F^{-1}(\omega), & \omega < \theta, \\ a, & \omega = \theta, \\ F^{-1}(\omega), & \omega > \theta. \end{cases}$$

The value of X at the single point $\omega = \theta$ is irrelevant for the distribution, since

$$\lambda(\{\theta\}) = 0.$$

Todo: Add the corresponding graph.

6.4.4 Case 4: General C.D.F.

Case 4.

For a general C.D.F., define the generalized inverse by

Definition 2. Generalized Inverse of a C.D.F.

For a C.D.F. F , its generalized inverse is defined by

$$F^{-1}(\omega) = \inf \{x \in \mathbb{R} : F(x) \geq \omega\}, \quad \omega \in (0, 1).$$

Define

$$X(\omega) = F^{-1}(\omega).$$

We need to show that

$$\mathbb{P}(X \leq b) = F(b)$$

and that X is measurable.

Let

$$A_b = \{\omega : X(\omega) \leq b\}.$$

It is enough to show that

$$A_b = (0, F(b)].$$

Claim. For every $b \in \mathbb{R}$,

$$\{\omega : F^{-1}(\omega) \leq b\} = (0, F(b)].$$

Consequently,

$$\begin{aligned}\mathbb{P}(X \leq b) &= \lambda(A_b) \\ &= \lambda((0, F(b)]) \\ &= F(b).\end{aligned}$$

Therefore,

$$F_X(b) = F(b)$$

for every $b \in \mathbb{R}$.

Homework: Prove the claim above and complete the remaining details of the construction.

End of Lecture

6.5 Quiz, 24 August 2026

Let $\mathcal{R}$ denote the respective appropriate Borel sigma algebra, while m is the Lebesgue measure on $(0, 1)$.

Quiz 3

1. Let $F : \mathbb{R} \rightarrow (0, 1)$ be a function such that it is non-decreasing, right continuous, and

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

Prove that there is a probability measure μ on $(\mathbb{R}, \mathcal{R})$ such that

$$F(x) = \mu((-\infty, x]).$$

2. Let $(\Omega, \mathcal{B}, \mu)$ be a probability space and X be a non-negative L^1 random variable. If F_X is the c.d.f., prove that

$$\int_{\mathbb{R}} (1 - F_X(x)) dx = \mathbb{E}(X).$$

3. **This is for Bonus:** Let μ be a probability measure on $(\mathbb{R}^2, \mathcal{R})$. Construct a random variable

$$X : ((0, 1), \mathcal{B}, m) \rightarrow \mathbb{R}^2$$

such that the distribution of X is μ .

Probability Theory

Lecture 07

Introduction to Probability

Date: Friday, 28 August 2026

Instructor: Dr. Nishant Chandgotia

7.1 Examples of Random Variables

Example 1. Uniform Random Variable

Let X be uniformly distributed on $(0, 1)$. Then its C.D.F. is given by

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

Example 2. Exponential Random Variable

Let

$$X \sim \text{Exp}(\lambda), \quad \lambda > 0.$$

Then X has density

$$f_X(y) = \lambda e^{-\lambda y}, \quad y \geq 0.$$

Example 3. Poisson Random Variable

Let

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0.$$

Then the C.D.F. of X is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \sum_{k=0}^{\lfloor x \rfloor} e^{-\lambda} \frac{\lambda^k}{k!}, & x \geq 0. \end{cases}$$

Example 4. Standard Normal Random Variable

Let

$$X \sim N(0, 1).$$

Then the C.D.F. of X is

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy.$$

7.2 A Gaussian Tail Estimate

Theorem 1. Gaussian Tail Bound

For $x > 0$,

$$\left(\frac{1}{x} - \frac{1}{x^3}\right) e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}.$$

Proof. We first prove the upper bound.

Let

$$y = x + z.$$

Then

$$\int_x^\infty e^{-y^2/2} dy = e^{-x^2/2} \int_0^\infty e^{-xz - z^2/2} dz.$$

Since

$$e^{-xz - z^2/2} \leq e^{-xz},$$

we obtain

$$\begin{aligned} \int_x^\infty e^{-y^2/2} dy &\leq e^{-x^2/2} \int_0^\infty e^{-xz} dz \\ &= \frac{1}{x} e^{-x^2/2}. \end{aligned}$$

Thus,

$$\boxed{\int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}}.$$

□

Todo: We have to compute the lower bound also.

7.3 Expectation and Variance

Definition 1. Expectation

Suppose X is a random variable such that

$$X \in L^1(\mathbb{P}).$$

Then the expectation of X is defined by

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P}.$$

Definition 2. Variance

Suppose

$$X \in L^2(\mathbb{P}).$$

Then the variance of X is defined by

$$\text{Var}(X) = \mathbb{E} \left[(X - \mathbb{E}(X))^2 \right].$$

7.4 Central Limit Theorem

Theorem 2. Central Limit Theorem

Let $X_1, X_2, \dots$ be i.i.d. random variables such that they all have same distribution say X , $\mathbb{E}(X_1^2) < \infty$. Let

$$\mu = \mathbb{E}(X_1), \quad \sigma^2 = \text{Var}(X_1).$$

Then

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{d} N(0, 1).$$

Equivalently, for $a < b$,

$$\mathbb{P} \left(a \leq \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \leq b \right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_a^b e^{-x^2/2} dx.$$

Remark

The result above is the usual form of the Central Limit Theorem.

7.5 Examples of Expectation and Variance

Example 5. Expectation and Variance of Common Distributions

1. If $X \sim \text{Exp}(\lambda)$, then

$$\mathbb{E}(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

2. If $X \sim \text{Poisson}(\lambda)$, then

$$\mathbb{E}(X) = \lambda, \quad \text{Var}(X) = \lambda.$$

3. If $X \sim \text{Geom}(p)$, then

$$\mathbb{E}(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}.$$

4. If $X \sim N(\mu, \sigma^2)$, then

$$\mathbb{E}(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

Question 1. Prove this

Do the calculations to prove $\mathbb{E}(X) = \mu$ and $\text{Var}(X) = \sigma^2$ for $X \sim N(\mu, \sigma^2)$.

Remark

If $Y = aX + b$, then $\mathbb{E}(Y) = a\mathbb{E}(X) + b$ and $\text{Var}(Y) = a^2 \text{Var}(X)$.

7.6 A Result on Signs of Vectors

Theorem 3. Sign Choice for Vectors

Let $v_1, \dots, v_n \in \mathbb{R}^n$ satisfy $\|v_i\|_2 = 1$ for all i .

Then:

1. There exist $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ such that $\|\sum_{i=1}^n \varepsilon_i v_i\|_2 \leq \sqrt{n}$.
2. There exist $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ such that $\|\sum_{i=1}^n \varepsilon_i v_i\|_2 \geq \sqrt{n}$.

Proof. Let $\varepsilon_1, \dots, \varepsilon_n$ be independent random variables, each uniformly distributed on $\{-1, 1\}$.

Then

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{i=1}^n \varepsilon_i v_i \right\|_2^2 \right] &= \mathbb{E} \left[\sum_{i,j} \varepsilon_i \varepsilon_j \langle v_i, v_j \rangle \right] \\ &= \sum_{i,j} \langle v_i, v_j \rangle \mathbb{E}(\varepsilon_i \varepsilon_j). \end{aligned}$$

For $i \neq j$, $\mathbb{E}(\varepsilon_i \varepsilon_j) = \mathbb{E}(\varepsilon_i) \mathbb{E}(\varepsilon_j) = 0$, while $\mathbb{E}(\varepsilon_i^2) = 1$.

Therefore,

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{i=1}^n \varepsilon_i v_i \right\|_2^2 \right] &= \sum_{i=1}^n \|v_i\|_2^2 \\ &= n. \end{aligned}$$

Hence there must exist a choice of signs for which $\|\sum_{i=1}^n \varepsilon_i v_i\|_2^2 \geq n$, and there must also exist a choice of signs for which $\|\sum_{i=1}^n \varepsilon_i v_i\|_2^2 \leq n$.

Taking square roots gives the result. □

7.7 Exercises on Probability Distributions

Question 2

Suppose X is a discrete random variable such that

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p, \quad k \geq 1, \quad k \in \mathbb{N}.$$

Find

$$\mathbb{P}(a < X < b).$$

Question 3

Suppose X is a random variable whose probability density function is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

Find

$$\mathbb{P}(a < X < b).$$

7.8 Graph Theory

Definition 3. Graph

A graph is a pair $G = (V, E)$, where

- V is the set of vertices;
- E is the set of edges.

In this lecture, we consider undirected graphs.

Example 6. Basic Graphs

Examples of graphs include:

1. The complete simple graph K_n .
2. The cycle graph C_n , where $V = \mathbb{Z}_n$ and $E = \{(i, i + 1) : i \in \mathbb{Z}_n\}$.

7.9 Bipartite Graphs

Definition 4. Bipartite Graph

A graph $G = (V, E)$ is called bipartite if there exists a partition $V = V_1 \sqcup V_2$ such that whenever $(u, v) \in E$, we have $u \in V_1, v \in V_2$, or equivalently, $u \in V_2, v \in V_1$.

Example 7. Odd Cycle is Not Bipartite

The cycle graph C_n is bipartite if and only if n is even.
For an odd cycle, one cannot partition the vertices into two sets such that every edge joins vertices belonging to different sets.

Example 8. Path Graph

The path graph P_n is always bipartite.

7.10 Subgraphs and Walks

Definition 5. Subgraph

A graph $G' = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

Definition 6. Walk

A walk from a vertex to another vertex in $G = (V, E)$ is a sequence of vertices $v_0, v_1, \dots, v_n$ such that $(v_i, v_{i+1}) \in E$ for every $i = 0, \dots, n-1$.

Definition 7. Tree

A graph is called a tree if it is connected and does not contain a cycle as a subgraph.

Definition 8. Connected Graph

A graph is called connected if, for all distinct vertices $u, v \in V$, there exists a walk from u to v .

7.11 Trees and Bipartite Graphs

Theorem 4. Every Tree is Bipartite

Every tree is a bipartite graph.

Proof. Let $T = (V, E)$ be a tree and choose a vertex $v \in V$.

Define

$$V_0 = \{u \in V : d(u, v) \text{ is even}\}$$

and

$$V_1 = \{u \in V : d(u, v) \text{ is odd}\},$$

where $d(u, v)$ denotes the length of the shortest walk from u to v .

Clearly,

$$V = V_0 \sqcup V_1.$$

Suppose $(u, w) \in E$. If $d(u, v)$ and $d(w, v)$ had the same parity, then combining shortest paths from v to u and v to w with the edge (u, w) would produce cycle.

Since T is a tree, this is impossible.

Therefore every edge joins a vertex in V_0 to a vertex in V_1 . Hence T is bipartite.

□

7.12 Characterization of Bipartite Graphs

Theorem 5. Characterization of Bipartite Graphs

A graph $G = (V, E)$ is bipartite if and only if it contains no cycle of odd length.

Remark

An odd cycle is not bipartite.

Moreover, every subgraph of a bipartite graph is also bipartite.

Proof. We have already seen that every tree is bipartite.

Now suppose that G is bipartite. Let $V = V_0 \sqcup V_1$ be a bipartition of V .

Consider a cycle

$$v_0, v_1, \dots, v_{n-1}, v_0.$$

Since every edge joins the two different parts of the bipartition, the vertices of the cycle must alternate between V_0 and V_1 .

Therefore the length of the cycle must be even.

Hence every cycle in a bipartite graph is even.

□

7.13 Bipartite Subgraph

Theorem 6. Bipartite Subgraph

Let $G = (V, E)$ be a finite graph without self-loops. Then there exists a bipartite subgraph $G' = (V, E')$ such that $|E'| \geq \frac{|E|}{2}$.

Proof. Let V be the vertex set and consider a random partition $V = V_0 \sqcup V_1$ obtained by independently assigning each vertex to V_0 or V_1 with probability $1/2$.

An edge $(u, v) \in E$ is said to be *crossing* if u and v belong to different parts of the partition.

Define the random variable

$$X_{(u,v)} = \begin{cases} 1, & \text{if } (u, v) \text{ is crossing,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $X = \sum_{(u,v) \in E} X_{(u,v)}$ be the total number of crossing edges.

For a fixed edge (u, v) ,

$$\begin{aligned} \mathbb{E}(X_{(u,v)}) &= \mathbb{P}(X_{(u,v)} = 1) \\ &= \mathbb{P}(u \in V_0, v \in V_1) + \mathbb{P}(u \in V_1, v \in V_0) \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}(X) &= \sum_{(u,v) \in E} \mathbb{E}(X_{(u,v)}) \\ &= \frac{|E|}{2}. \end{aligned}$$

Hence there must exist at least one partition for which

$$X \geq \frac{|E|}{2}.$$

Take E' to be the set of crossing edges for such a partition. Then $G' = (V, E')$ is bipartite and satisfies $|E'| \geq \frac{|E|}{2}$.

□

End of Lecture

Probability Theory

Lecture 08

Introduction to Probability

Date: Wednesday, 2 September 2026

Instructor: Dr. Nishant Chandgotia

Suppose $X, Y \in L^1(\mathbb{P})$ and X and Y are independent. Then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.

Question 1. Why does this make sense?

Why does $XY \in L^1(\mathbb{P})$?

Remark

The basic idea is that if $X \sim \mu$ and $Y \sim \nu$ are independent, then

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B) = \mu(A)\nu(B).$$

Hence, $(X, Y) \sim \mu \times \nu$.

Therefore,

$$\begin{aligned}\mathbb{E}(|XY|) &= \mathbb{E}(|X||Y|) \\ &= \int_{\mathbb{R}^2} |x||y| d(\mu \times \nu)(x, y).\end{aligned}$$

By Fubini's theorem,

$$\begin{aligned}\mathbb{E}(|XY|) &= \left(\int_{\mathbb{R}} |x| d\mu(x) \right) \left(\int_{\mathbb{R}} |y| d\nu(y) \right) \\ &= \mathbb{E}(|X|)\mathbb{E}(|Y|) < \infty.\end{aligned}$$

The same argument applied to XY gives

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y).$$

This also works when $X, Y \geq 0$.

Similarly, if $X_1, X_2, \dots, X_n \in L^1(\mathbb{P})$ are independent, then

$$\mathbb{E}\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n \mathbb{E}(X_i).$$

Definition 1. Covariance

For random variables $X, Y \in L^2(\mathbb{P})$, the covariance of X and Y is defined by

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))].$$

Equivalently,

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Remark

If X and Y are independent, then $\text{Cov}(X, Y) = 0$.
The converse is not true in general.

Example 1. Uncorrelated but Dependent Random Variables

Consider random variables X, Y taking values in $\{-1, 0, 1\}$ with joint probability table

	$Y = 1$	$Y = 0$	$Y = -1$
$X = 1$	0	a	0
$X = 0$	b	c	b
$X = -1$	0	a	0

where $2a + 2b + c = 1$, $a, b, c \geq 0$.

Clearly this is a probability distribution.

We have $\mathbb{P}(X = 1, Y = 1) = 0$, but $\mathbb{P}(X = 1)\mathbb{P}(Y = 1) > 0$ whenever $a, b > 0$. Hence X and Y are not independent.

On the other hand, $\mathbb{E}(X) = 0$, $\mathbb{E}(Y) = 0$,
and therefore

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 0.$$

Thus zero covariance does not imply independence.

Question 2. Distribution of a Transformation

If $X \sim \text{Unif}[-1, 1]$, what is the distribution of X^3 ?

Question 3. Bernoulli Random Variables

Suppose $X, Y \sim \text{Ber}$.

What conditions on X and Y imply independence?

Question 4. Jointly Normal Random Variables

Suppose X and Y are jointly normal random variables.

Does $\text{Cov}(X, Y) = 0$ imply that X and Y are independent?

Question 5. Bernoulli Variables and Covariance

Suppose $X \sim \text{Ber}(p)$, $Y \sim \text{Ber}(q)$.

If $\text{Cov}(X, Y) = 0$, does it follow that X and Y are independent?

8.1 Inequalities

Definition 2. Markov's Inequality

Let $X \geq 0$ be a random variable and let $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}(X)}{a}.$$

Remark

The main idea behind Markov's inequality is the following. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function such that $\varphi \geq 0$.

For $A \in \mathcal{F}$,

$$\mathbb{E}[\varphi(X)\mathbf{1}_{\{X \in A\}}] \leq \mathbb{E}[\varphi(X)].$$

This simple inequality has many important consequences.

Definition 3. Chebyshev's Inequality

Let $\mu = \mathbb{E}(X)$, $\sigma^2 = \text{Var}(X)$, $\lambda > 0$.

Then

$$\mathbb{P}(|X - \mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2}.$$

Remark

Let, $Y = (X - \mu)^2$. Then $\mathbb{E}(Y) = \text{Var}(X) = \sigma^2$.

Therefore, by Markov's inequality,

$$\begin{aligned} \mathbb{P}(|X - \mu| \geq \lambda\sigma) &= \mathbb{P}(Y \geq \lambda^2\sigma^2) \\ &\leq \frac{\sigma^2}{\lambda^2\sigma^2} \\ &= \frac{1}{\lambda^2}. \end{aligned}$$

Definition 4. Chernoff Bound

For $t > 0$, define the moment generating function

$$M(t) = \mathbb{E}(e^{tX}).$$

Then

$$\mathbb{P}(X \geq a) \leq \frac{M(t)}{e^{ta}}.$$

Remark

The Chernoff bound follows immediately by applying Markov's inequality to the nonnegative random variable e^{tX} :

$$\begin{aligned}\mathbb{P}(X \geq a) &= \mathbb{P}(e^{tX} \geq e^{ta}) \\ &\leq \frac{\mathbb{E}(e^{tX})}{e^{ta}} \\ &= \frac{M(t)}{e^{ta}}.\end{aligned}$$

8.2 Coupon Collector Problem

Question 6. Coupon Collector Problem

Suppose there are n different types of cards. Each time a card is picked, each of the n types is obtained independently and uniformly.

How many cards do we need to draw to ensure that we have obtained all n types of cards?

Consider the probability space $\Omega = (\{1, 2, \dots, n\}^{\mathbb{N}}, \mathcal{B}, \text{Unif}^{\otimes \mathbb{N}})$.

Let $X_n(\omega) = \#\{\text{cards picked to obtain the full collection}\}$.

Equivalently,

$$X_n(\omega) = \min \{m : \{\omega_1, \omega_2, \dots, \omega_m\} = \{1, \dots, n\}\}.$$

Theorem 1. Coupon Collector Bound

For every $\varepsilon > 0$,

$$\mathbb{P}\left(1 - \varepsilon < \frac{X_n}{n \log n} < 1 + \varepsilon\right) \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

Proof. We prove the upper bound. It is enough to show that

$$\mathbb{P}\left(\frac{X_n}{n \log n} > 1 + \varepsilon\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Equivalently,

$$\mathbb{P}(X_n > (1 + \varepsilon)n \log n) \rightarrow 0.$$

Let Y_k denote the number of card types missing after k draws. Thus,

$$Y_k(\omega) = n - |\{\omega_1, \omega_2, \dots, \omega_k\}|.$$

We want to show that if $k > (1 + \varepsilon)n \log n$, then Y_k is very likely to be zero.

For each $i \in \{1, \dots, n\}$ and each $k \geq 1$, define

$$S_{i,k} = \begin{cases} 1, & \text{if card type } i \text{ is not picked by time } k, \\ 0, & \text{otherwise.} \end{cases}$$

Then $Y_k = \sum_{i=1}^n S_{i,k}$.

For a fixed i , $\mathbb{E}(S_{i,k}) = \left(1 - \frac{1}{n}\right)^k$, since the probability that type i is not selected in one draw is $1 - \frac{1}{n}$.

Therefore,

$$\mathbb{E}(Y_k) = n \left(1 - \frac{1}{n}\right)^k.$$

Now take $k = (1 + \varepsilon)n \log n$.

By Markov's inequality,

$$\begin{aligned} \mathbb{P}(X_n > (1 + \varepsilon)n \log n) &= \mathbb{P}(Y_{(1+\varepsilon)n \log n} \geq 1) \\ &\leq \mathbb{E}[Y_{(1+\varepsilon)n \log n}] \\ &= n \left(1 - \frac{1}{n}\right)^{(1+\varepsilon)n \log n}. \end{aligned}$$

Using $1 - \frac{1}{n} \leq e^{-1/n}$, we obtain

$$\begin{aligned} n \left(1 - \frac{1}{n}\right)^{(1+\varepsilon)n \log n} &\leq n \exp\left(-\frac{(1 + \varepsilon)n \log n}{n}\right) \\ &= n e^{-(1+\varepsilon) \log n} \\ &= n^{-\varepsilon}. \end{aligned}$$

Hence,

$$\mathbb{P}(X_n > (1 + \varepsilon)n \log n) \leq n^{-\varepsilon} \longrightarrow 0.$$

Thus,

$$\mathbb{P}\left(\frac{X_n}{n \log n} > 1 + \varepsilon\right) \longrightarrow 0.$$

This proves the desired upper bound. □

8.3 Quiz, 4 September 2026

Quiz 4

1. It is equivalent to a question discussed in class. What is it?
2. If $X \sim \text{Geo}(p)$, prove that $\mathbb{E}(X) = \frac{1}{p}$, $\text{Var}(X) \leq \frac{1}{p^2}$.
3. Let $X_k^{(n)}$ be the time taken for k bins to be filled. Prove that $S_k^{(n)} = X_{k+1}^{(n)} - X_k^{(n)} \sim \text{Geo}(p_k)$, where $p_k = 1 - \frac{k}{n}$, and that the random variables $S_0^{(n)}, S_1^{(n)}, \dots, S_{n-1}^{(n)}$ are independent.
4. Prove that $\mathbb{E}(X_n^{(n)}) = n \sum_{m=1}^n \frac{1}{m}$.
5. Prove that $\text{Var}(X_n^{(n)}) \leq cn^2$ for some constant c .
6. Prove that $\frac{\mathbb{E}(X_n^{(n)})}{n \log n} \rightarrow 1$ as $n \rightarrow \infty$.
7. Use Chebyshev's inequality to conclude that $\frac{X_n^{(n)} - n \sum_{m=1}^n \frac{1}{m}}{nf(n)} \rightarrow 0$ in probability as $f(n) \rightarrow \infty$.

Probability Theory

Lecture 09

Introduction to Probability

Date: Monday, 7 September 2026

Instructor: Dr. Nishant Chandgotia

9.1 Continuation of Coupon Collector Problem

We have already defined X_n and $Y_k = \sum_{i=1}^n S_{i,k}$.

Now,

$$\text{Var}(Y_k) = \sum_{i=1}^n \text{Var}(S_{i,k}) + \sum_{\substack{i,j=1 \\ i \neq j}}^n \text{Cov}(S_{i,k}, S_{j,k}).$$

We have

$$\text{Var}(S_{i,k}) = \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right].$$

Now,

$$\begin{aligned} \text{Cov}(S_{i,k}, S_{j,k}) &= \mathbb{E}(S_{i,k} S_{j,k}) - \mathbb{E}(S_{i,k}) \mathbb{E}(S_{j,k}) \\ &= \left(1 - \frac{2}{n}\right)^k - \left(1 - \frac{1}{n}\right)^{2k}. \end{aligned}$$

Since

$$(1 - x)^2 \geq 1 - 2x,$$

we have

$$\left(1 - \frac{1}{n}\right)^2 \geq 1 - \frac{2}{n},$$

and hence

$$\left(1 - \frac{2}{n}\right)^k \leq \left(1 - \frac{1}{n}\right)^{2k}.$$

Therefore,

$$\text{Cov}(S_{i,k}, S_{j,k}) \leq 0.$$

Thus,

$$\text{Var}(Y_k) \leq n \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right].$$

Also,

$$\mathbb{E}(Y_k) = n \left(1 - \frac{1}{n}\right)^k.$$

Therefore,

$$\begin{aligned} \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} &\geq \frac{n \left(1 - \frac{1}{n}\right)^k}{\sqrt{n \left(1 - \frac{1}{n}\right)^k \left[1 - \left(1 - \frac{1}{n}\right)^k\right]}} \\ &\geq \sqrt{n \left(1 - \frac{1}{n}\right)^k}. \end{aligned}$$

Question 1. Coupon Collector Problem

We want to show that

$$\mathbb{P}(X_n < (1 - \epsilon)n \log n) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Consider

$$k = (1 - \epsilon)n \log n.$$

Then

$$\mathbb{P}(X_n < k) = \mathbb{P}(Y_k = 0).$$

By Chebyshev's inequality,

$$\begin{aligned} \mathbb{P}(Y_k = 0) &\leq \mathbb{P}\left(\left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| \geq \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right) \\ &\leq \frac{\text{Var}(Y_k)}{\mathbb{E}(Y_k)^2}. \end{aligned}$$

More generally, Chebyshev gives

$$\mathbb{P}\left(\left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| \geq t\right) \leq \frac{1}{t^2}.$$

We have

$$|Y_k - \mathbb{E}(Y_k)| \geq t\sqrt{\text{Var}(Y_k)} \iff Y_k \notin \left(\mathbb{E}(Y_k) - t\sqrt{\text{Var}(Y_k)}, \mathbb{E}(Y_k) + t\sqrt{\text{Var}(Y_k)}\right).$$

Choose t such that

$$0 < \mathbb{E}(Y_k) - t\sqrt{\text{Var}(Y_k)}.$$

Thus,

$$t \leq \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}.$$

If

$$Y_k = 0,$$

then

$$\begin{aligned} \left|\frac{Y_k - \mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}}\right| &= \frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} \\ &\geq \sqrt{n \left(1 - \frac{1}{n}\right)^k}. \end{aligned}$$

For

$$k = (1 - \epsilon)n \log n,$$

we have

$$\begin{aligned} n \left(1 - \frac{1}{n}\right)^k &\sim n e^{-(1-\epsilon) \log n} \\ &= n^\epsilon. \end{aligned}$$

Hence,

$$\frac{\mathbb{E}(Y_k)}{\sqrt{\text{Var}(Y_k)}} \geq \sqrt{n^\epsilon} = n^{\epsilon/2}.$$

Therefore,

$$\mathbb{P}(Y_{(1-\epsilon)n \log n} = 0) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Todo: Check and fill the corresponding graphs.

9.2 Erdős–Rényi Graphs

Let $n \in \mathbb{N}$, $p \in (0, 1)$. The random graph $G(n, p)$ is a random graph with n vertices $\{1, 2, \dots, n\}$, where each edge is chosen independently with probability p .

Question 2. Number of Copies of K_r

How many copies of K_r are there in $G(n, p)$, where K_r denotes the complete graph on r vertices?

Let $A \subseteq \{1, 2, \dots, n\}$ have cardinality r .

Define $X_A = \begin{cases} 1, & \text{if } K_r \text{ appears on } A, \\ 0, & \text{otherwise.} \end{cases}$

Let X denote the number of copies of K_r . Then

$$X = \sum_{\substack{A \subseteq \{1, 2, \dots, n\} \\ |A|=r}} X_A.$$

Since all the edges of K_r must be present,

$$\mathbb{E}(X_A) = p^{\binom{r}{2}}.$$

Therefore,

$$\begin{aligned} \mathbb{E}(X) &= \sum_{\substack{A \subseteq \{1, 2, \dots, n\} \\ |A|=r}} \mathbb{E}(X_A) \\ &= \binom{n}{r} p^{\binom{r}{2}}. \end{aligned}$$

Now,

$$\begin{aligned}\text{Var}(X) &= \sum_{\substack{A \subseteq \{1,2,\dots,n\} \\ |A|=r}} \text{Var}(X_A) \\ &\quad + \sum_{\substack{A \neq B \\ |A|=|B|=r}} \text{Cov}(X_A, X_B).\end{aligned}$$

For the first term,

$$\sum_{|A|=r} \text{Var}(X_A) = \binom{n}{r} p^{\binom{r}{2}} (1 - p^{\binom{r}{2}}).$$

If $|A \cap B| \leq 1$, then the sets of edges involved in X_A and X_B are disjoint.

Hence,

$$\text{Cov}(X_A, X_B) = 0.$$

If $|A \cap B| = t > 1$, then

$$\begin{aligned}\text{Cov}(X_A, X_B) &= \mathbb{E}(X_A X_B) - \mathbb{E}(X_A) \mathbb{E}(X_B) \\ &= p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}}.\end{aligned}$$

Thus,

$$\sum_{\substack{A \neq B, |A \cap B|=t \\ |A|=|B|=r}} \text{Cov}(X_A, X_B) = \binom{n}{r} \binom{n-r}{r-t} [p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}}].$$

Consequently,

$$\begin{aligned}\text{Var}(X) &= \binom{n}{r} p^{\binom{r}{2}} (1 - p^{\binom{r}{2}}) \\ &\quad + \sum_{t=2}^{r-1} \binom{n}{r} \binom{n-r}{r-t} [p^{2\binom{r}{2} - \binom{t}{2}} - p^{2\binom{r}{2}}].\end{aligned}$$

For some constants $c_1, c_2 > 0$,

$$\mathbb{E}(X) \sim c_1 n^r, \quad \text{Var}(X) \sim c_2 n^{2r-2}.$$

By Chebyshev's inequality, we have

$$\mathbb{P}\left(\left|\frac{X - \mathbb{E}(X)}{n^{r-1} f(n)}\right| > \epsilon\right) \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

where $f(n)$ is any function such that $f(n) \longrightarrow \infty$ as $n \rightarrow \infty$.

In fact,

$$\begin{aligned}\mathbb{P}(|X - \mathbb{E}(X)| > \epsilon \mathbb{E}(X)) &\leq \frac{\text{Var}(X)}{\epsilon^2 \mathbb{E}(X)^2} \\ &\sim \frac{1}{n^2 \epsilon^2} \longrightarrow 0.\end{aligned}$$

9.2.1 Full Strength of Chebyshev

Let $X_1, X_2, \dots, X_n$ be random variables and define $S_n = \sum_{i=1}^n X_i$.

Suppose $\mathbb{E}(X_i) = \mu$, $\text{Var}(X_i) \leq C$ for all i , and

$$\text{Cov}(X_i, X_j) = 0 \quad \text{for all } i < j.$$

Thus, the random variables are uncorrelated.

Then $\frac{S_n}{n} \rightarrow \mu$ in L^2 and in probability.

This is a version of the weak law of large numbers.

Proof.

We have

$$\begin{aligned} \mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right|^2 \right] &= \text{Var} \left(\frac{S_n}{n} \right) \\ &= \frac{1}{n^2} \text{Var}(S_n). \end{aligned}$$

Since the random variables are uncorrelated,

$$\begin{aligned} \text{Var}(S_n) &= \sum_{i=1}^n \text{Var}(X_i) \\ &\leq Cn. \end{aligned}$$

Therefore,

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right|^2 \right] \leq \frac{Cn}{n^2} = \frac{C}{n} \rightarrow 0.$$

Hence, $\frac{S_n}{n} \rightarrow \mu$ in L^2 , and therefore $\frac{S_n}{n} \rightarrow \mu$ in probability.

But further:

By Chebyshev's inequality,

$$\begin{aligned} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \epsilon \mu \right) &\leq \frac{\text{Var}(S_n)}{(n\epsilon\mu)^2} \\ &\leq \frac{nC}{n^2\epsilon^2\mu^2} \\ &= \frac{C}{n\epsilon^2\mu^2} \rightarrow 0. \end{aligned}$$

Question 3. Exponential Decay

How do we get exponential decay?

End of Lecture

9.3 Quiz, 09 September 2026

Quiz 5

1. Let X be a non-negative integer-valued random variable with finite variance. Prove that

$$\mathbb{P}(X = 0) \leq \frac{\text{Var}(X)}{\mathbb{E}(X^2)}.$$

2. Let X be a random variable with $\mathbb{E}(X) = 0$ and $\text{Var}(X) = \sigma^2$. Prove that for all $\lambda > 0$,

$$\mathbb{P}(X \geq \lambda) \leq \frac{\sigma^2}{\lambda^2 + \sigma^2}.$$

Probability Theory

Lecture 10

Introduction to Probability

Date: Wednesday, 11 September 2026

Instructor: Dr. Nishant Chandgotia

10.1 Hoeffding's Inequality

Theorem 1. Hoeffding's Inequality

Let $X_1, X_2, \dots, X_n$ be independent random variables such that $\mathbb{E}(X_i) = 0$ for all i , and $|X_i| \leq a_i$.

Define

$$S_n = \sum_{i=1}^n X_i.$$

Then, for every $t > 0$,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right).$$

Corollary 1. Two-Sided Hoeffding Bound

Under the assumptions of Hoeffding's inequality,

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right), \quad t > 0.$$

Remark

There are several useful versions of Hoeffding's inequality.

1. The inequality can be applied to centered versions of random variables. If the random variables do not have mean zero, replace S_n by

$$S_n - \sum_{i=1}^n \mathbb{E}(X_i).$$

2. There are versions of the inequality for asymmetric intervals.
3. Independence can be relaxed in other concentration inequalities. One important example is the *Azuma-Hoeffding inequality* for martingales.

Example 1. Coin Tosses

Consider independent coin tosses and let

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th toss is a head,} \\ 0, & \text{if the } i\text{-th toss is a tail.} \end{cases}$$

After centering the variables appropriately, Hoeffding's inequality gives an exponential bound on deviations of the number of heads from its expected value.

In particular, when the variables satisfy

$$|X_i| \leq 1,$$

we obtain a bound of the form

$$\mathbb{P}(|S_n| \geq t) \leq 2 \exp\left(-\frac{t^2}{2n}\right).$$

For $t = n\alpha$,

$$\mathbb{P}(|S_n| \geq n\alpha) \leq 2 \exp\left(-\frac{n\alpha^2}{2}\right).$$

Thus the probability of a deviation of order n decays exponentially in n .

Proof. For $\theta > 0$, by Markov's inequality,

$$\begin{aligned} \mathbb{P}(S_n \geq t) &= \mathbb{P}\left(e^{\theta S_n} \geq e^{\theta t}\right) \\ &\leq \frac{\mathbb{E}\left(e^{\theta S_n}\right)}{e^{\theta t}}. \end{aligned}$$

Since the X_i are independent,

$$\mathbb{E}\left(e^{\theta S_n}\right) = \prod_{i=1}^n \mathbb{E}\left(e^{\theta X_i}\right).$$

Because $-a_i \leq X_i \leq a_i$ and $\mathbb{E}(X_i) = 0$, convexity of the exponential function gives

$$\mathbb{E}\left(e^{\theta X_i}\right) \leq \frac{1}{2}\left(e^{-\theta a_i} + e^{\theta a_i}\right).$$

Using the power-series expansion of the exponential function,

$$\frac{e^{-u} + e^u}{2} \leq e^{u^2/2},$$

and hence

$$\mathbb{E}\left(e^{\theta X_i}\right) \leq e^{\theta^2 a_i^2/2}.$$

Therefore,

$$\begin{aligned} \mathbb{P}(S_n \geq t) &\leq \frac{\exp\left(\frac{\theta^2}{2} \sum_{i=1}^n a_i^2\right)}{e^{\theta t}} \\ &= \exp\left(\frac{\theta^2}{2} \sum_{i=1}^n a_i^2 - \theta t\right). \end{aligned}$$

Choose $\theta = \frac{t}{\sum_{i=1}^n a_i^2}$.

Then

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n a_i^2}\right).$$

This proves Hoeffding's inequality.

□

10.2 Independence

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let I be an index set.

Definition 1. Independence of Random Variables

Let $X_i : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$, $i \in I$, be random variables.

The collection $(X_i)_{i \in I}$ is said to be independent if, for every finite collection of distinct indices $i_1, i_2, \dots, i_n \in I$ and measurable sets $A_1, A_2, \dots, A_n \in \mathcal{S}$, we have

$$\mathbb{P}\left(\bigcap_{j=1}^n \{X_{i_j} \in A_j\}\right) = \prod_{j=1}^n \mathbb{P}(X_{i_j} \in A_j).$$

Definition 2. Independence of σ -Algebras

Let $\{\mathcal{F}_i\}_{i \in I}$ be a collection of sub- σ -algebras of $\mathcal{F}$.

They are said to be independent if, for every finite collection of distinct indices $i_1, i_2, \dots, i_n \in I$ and sets $B_{i_j} \in \mathcal{F}_{i_j}$, we have

$$\mathbb{P}\left(\bigcap_{j=1}^n B_{i_j}\right) = \prod_{j=1}^n \mathbb{P}(B_{i_j}).$$

Definition 3. σ -Algebra Generated by a Random Variable

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (S, \mathcal{S})$ be a random variable.

The σ -algebra generated by X is

$$\sigma(X) = \sigma\left(X^{-1}(A) : A \in \mathcal{S}\right).$$

Equivalently, $\sigma(X)$ is the smallest σ -algebra contained in $\mathcal{F}$ with respect to which X is measurable.

Observation

The random variables $(X_i)_{i \in I}$ are independent if and only if the σ -algebras $(\sigma(X_i))_{i \in I}$ are independent.

Proposition 1. Basic Properties of Independence

Let $B, B' \in \mathcal{F}$.

Then

$$B \perp B' \iff B \perp (B')^c \iff B^c \perp B'.$$

Proposition 2. Disjoint Independent Events

Let $B_1, B_2, \dots, B_n, B \in \mathcal{F}$.

Suppose that the events $B_1, \dots, B_n$ are pairwise disjoint and $B_i \perp B$ for every i .

Then

$$\bigcup_{i=1}^n B_i \perp B.$$

Question 1. Intersection of Independent Events

Suppose

$$B_1 \perp B \quad \text{and} \quad B_2 \perp B.$$

Does it follow that

$$B_1 \cap B_2 \perp B?$$

TODO

Find an explicit counterexample, or use a Venn diagram to determine whether the statement is true in general.

10.3 π - λ Systems

A useful way of checking independence is to reduce the collection of sets that needs to be checked.

Definition 4. π -System

Let

$$\mathcal{P} \subseteq \mathcal{F}.$$

We say that $\mathcal{P}$ is a π -system if it is closed under finite intersections; that is,

$$A, B \in \mathcal{P} \implies A \cap B \in \mathcal{P}.$$

Definition 5. λ -System

Let

$$\mathcal{L} \subseteq \mathcal{F}.$$

We say that $\mathcal{L}$ is a λ -system if:

1. $\Omega \in \mathcal{L}$.
2. If $A, B \in \mathcal{L}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{L}$.
3. If $A_1 \subseteq A_2 \subseteq \dots, \subseteq A_n, \dots \in \mathcal{L}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{L}$.

Theorem 2. π - λ Theorem

Let $\mathcal{P}$ be a π -system and let $\mathcal{L}$ be a λ -system such that $\mathcal{P} \subseteq \mathcal{L}$. Then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.

Proof. Proof Sketch: Durrett Let $\mathcal{L}'$ be the smallest λ -system containing $\mathcal{P}$. Thus,

$$\mathcal{P} \subseteq \mathcal{L}' \subseteq \mathcal{L}.$$

The π - λ theorem states that if $\mathcal{P}$ is a π -system, then

$$\sigma(\mathcal{P}) = \mathcal{L}'.$$

Therefore,

$$\sigma(\mathcal{P}) \subseteq \mathcal{L}.$$

□

Remark

The π - λ theorem allows us to verify independence on a smaller collection of sets. Instead of checking independence for every set in

$$\sigma(X_i),$$

it is often enough to check it on a suitable π -system generating that σ -algebra.

End of Lecture

Probability Theory

Lecture 11

Introduction to Probability

Date: Wednesday, 16 September 2026

Instructor: Dr. Nishant Chandgotia

Theorem 1. Independence of Generated σ -Fields

Suppose that

$$\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$$

are independent π -systems. Then

$$\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$$

are independent.

Proof. We first prove the following intermediate statement:

(*) If $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent, then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent.

Once (*) is proved, we can apply the same argument successively to $\mathcal{A}_2, \dots, \mathcal{A}_n$.

Fix

$$A_i \in \mathcal{A}_i, \quad 2 \leq i \leq n,$$

and define

$$F = A_2 \cap A_3 \cap \dots \cap A_n.$$

Define

$$\mathcal{L} = \{A : \mathbb{P}(A \cap F) = \mathbb{P}(A)\mathbb{P}(F)\}.$$

We will show that $\mathcal{L}$ is a λ -system and that

$$\mathcal{A}_1 \subseteq \mathcal{L}.$$

Then the π - λ theorem will imply

$$\sigma(\mathcal{A}_1) \subseteq \mathcal{L}.$$

Step 1: $\Omega \in \mathcal{L}$.

Since

$$\Omega \cap F = F$$

and

$$\mathbb{P}(\Omega) = 1,$$

we have

$$\mathbb{P}(\Omega \cap F) = \mathbb{P}(F) = \mathbb{P}(\Omega)\mathbb{P}(F).$$

Hence

$$\Omega \in \mathcal{L}.$$

Step 2: Difference property.

Suppose

$$A, B \in \mathcal{L}, \quad A \subseteq B.$$

Then

$$(B \setminus A) \cap F = (B \cap F) \setminus (A \cap F).$$

Since

$$A \cap F \subseteq B \cap F,$$

the difference formula for a probability measure gives

$$\mathbb{P}((B \setminus A) \cap F) = \mathbb{P}(B \cap F) - \mathbb{P}(A \cap F).$$

Since $A, B \in \mathcal{L}$,

$$\mathbb{P}(B \cap F) = \mathbb{P}(B)\mathbb{P}(F)$$

and

$$\mathbb{P}(A \cap F) = \mathbb{P}(A)\mathbb{P}(F).$$

Therefore

$$\begin{aligned} \mathbb{P}((B \setminus A) \cap F) &= \mathbb{P}(B)\mathbb{P}(F) - \mathbb{P}(A)\mathbb{P}(F) \\ &= (\mathbb{P}(B) - \mathbb{P}(A))\mathbb{P}(F). \end{aligned}$$

Since $A \subseteq B$,

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A).$$

Hence

$$\mathbb{P}((B \setminus A) \cap F) = \mathbb{P}(B \setminus A)\mathbb{P}(F).$$

Therefore

$$B \setminus A \in \mathcal{L}.$$

Step 3: Continuity from below.

Suppose

$$B_k \in \mathcal{L}, \quad B_k \uparrow B.$$

Then

$$B_k \cap F \uparrow B \cap F.$$

Indeed,

$$B_k \subseteq B_{k+1} \implies B_k \cap F \subseteq B_{k+1} \cap F,$$

and

$$\begin{aligned}\bigcup_{k=1}^{\infty} (B_k \cap F) &= \left(\bigcup_{k=1}^{\infty} B_k \right) \cap F \\ &= B \cap F.\end{aligned}$$

By continuity from below,

$$\mathbb{P}(B \cap F) = \lim_{k \rightarrow \infty} \mathbb{P}(B_k \cap F).$$

Since $B_k \in \mathcal{L}$,

$$\mathbb{P}(B_k \cap F) = \mathbb{P}(B_k)\mathbb{P}(F).$$

Therefore

$$\begin{aligned}\mathbb{P}(B \cap F) &= \lim_{k \rightarrow \infty} \mathbb{P}(B_k)\mathbb{P}(F) \\ &= \mathbb{P}(F) \lim_{k \rightarrow \infty} \mathbb{P}(B_k).\end{aligned}$$

Again, by continuity from below,

$$\lim_{k \rightarrow \infty} \mathbb{P}(B_k) = \mathbb{P}(B).$$

Consequently,

$$\mathbb{P}(B \cap F) = \mathbb{P}(B)\mathbb{P}(F).$$

Hence

$$B \in \mathcal{L}.$$

Thus $\mathcal{L}$ is a λ -system.

Step 4: Show that $\mathcal{A}_1 \subseteq \mathcal{L}$.

Take

$$A_1 \in \mathcal{A}_1.$$

Since

$$A_i \in \mathcal{A}_i, \quad 1 \leq i \leq n,$$

and

$$\mathcal{A}_1, \dots, \mathcal{A}_n$$

are independent, we have

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i).$$

Also, since

$$\mathcal{A}_2, \dots, \mathcal{A}_n$$

are independent,

$$\mathbb{P}(F) = \mathbb{P}(A_2 \cap \dots \cap A_n) = \prod_{i=2}^n \mathbb{P}(A_i).$$

Therefore

$$\begin{aligned}
 \mathbb{P}(A_1 \cap F) &= \mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) \\
 &= \prod_{i=1}^n \mathbb{P}(A_i) \\
 &= \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}(A_i) \\
 &= \mathbb{P}(A_1) \mathbb{P}(F).
 \end{aligned}$$

Thus

$$A_1 \in \mathcal{L}.$$

Since A_1 was arbitrary,

$$\mathcal{A}_1 \subseteq \mathcal{L}.$$

Step 5: Apply the π - λ theorem.

The collection $\mathcal{A}_1$ is a π -system and $\mathcal{L}$ is a λ -system containing $\mathcal{A}_1$.

Therefore, by the π - λ theorem,

$$\sigma(\mathcal{A}_1) \subseteq \mathcal{L}.$$

Hence, for every

$$A_1 \in \sigma(\mathcal{A}_1),$$

we have

$$\mathbb{P}(A_1 \cap F) = \mathbb{P}(A_1) \mathbb{P}(F).$$

Since

$$F = A_2 \cap \cdots \cap A_n,$$

it follows that

$$\begin{aligned}
 \mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) &= \mathbb{P}(A_1) \mathbb{P}(A_2 \cap \cdots \cap A_n) \\
 &= \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}(A_i) \\
 &= \prod_{i=1}^n \mathbb{P}(A_i).
 \end{aligned}$$

Thus

$$\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$$

are independent.

This proves (*).

Step 6: Iterate the argument.

We have shown that

$$\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$$

are independent.

By permuting the order of the systems, we may apply (*) again and obtain

$$\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \mathcal{A}_3, \dots, \mathcal{A}_n$$

as independent.

Repeating this argument successively gives

$$\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$$

as independent.

Therefore

$$\sigma(\mathcal{A}_1), \dots, \sigma(\mathcal{A}_n) \text{ are independent.}$$

□

Observation

A π -System Criterion for Independence Let

$$X_1, X_2 : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (S, \mathcal{R})$$

be random variables.

Suppose that $\mathcal{A} \subseteq \mathcal{R}$ is a π -system such that $\sigma(\mathcal{A}) = \mathcal{R}$.

Then, to prove that X_1 and X_2 are independent, it is enough to verify that

$$\mathbb{P}(X_1 \in A_1, X_2 \in A_2) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)$$

for all $A_1, A_2 \in \mathcal{A}$.

Indeed,

$$\mathcal{A}$$

generates the relevant σ -field, and the preceding π - λ argument extends the independence from $\mathcal{A}$ to $\sigma(\mathcal{A})$.

In particular, when

$$(S, \mathcal{R}) = (\mathbb{R}, \mathcal{B}(\mathbb{R})),$$

we may take

$$\mathcal{A} = \{(-\infty, a] : a \in \mathbb{R}\},$$

which is a π -system generating $\mathcal{B}(\mathbb{R})$.

Theorem 2. Independence of Groups of Independent σ -Fields

Suppose that

$$\mathcal{F}_{i,j}, \quad 1 \leq i \leq n, \quad 1 \leq j \leq m(i),$$

are independent σ -fields, and let

$$\mathcal{G}_i = \sigma \left(\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j} \right).$$

Then $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent.

Proof. For each i , define

$$\mathcal{A}_i = \left\{ \bigcap_{j=1}^{m(i)} A_{i,j} : A_{i,j} \in \mathcal{F}_{i,j} \right\}.$$

Since each $\mathcal{F}_{i,j}$ is a σ -field, in particular

$$\Omega \in \mathcal{F}_{i,j}.$$

We first show that $\mathcal{A}_i$ is a π -system.

Take

$$A_i = \bigcap_{j=1}^{m(i)} A_{i,j}, \quad B_i = \bigcap_{j=1}^{m(i)} B_{i,j},$$

where

$$A_{i,j}, B_{i,j} \in \mathcal{F}_{i,j}.$$

Then

$$A_i \cap B_i = \bigcap_{j=1}^{m(i)} (A_{i,j} \cap B_{i,j}).$$

Since each $\mathcal{F}_{i,j}$ is a σ -field,

$$A_{i,j} \cap B_{i,j} \in \mathcal{F}_{i,j}.$$

Therefore

$$A_i \cap B_i \in \mathcal{A}_i.$$

Hence $\mathcal{A}_i$ is a π -system.

Moreover, for every

$$A_{i,j} \in \mathcal{F}_{i,j},$$

we have

$$A_{i,j} = A_{i,j} \cap \bigcap_{k \neq j} \Omega.$$

Since

$$\Omega \in \mathcal{F}_{i,k},$$

it follows that

$$A_{i,j} \in \mathcal{A}_i.$$

Thus

$$\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j} \subseteq \mathcal{A}_i.$$

On the other hand, every element of $\mathcal{A}_i$ is a finite intersection of sets belonging to

$$\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j}.$$

Hence

$$\mathcal{A}_i \subseteq \sigma \left(\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j} \right) = \mathcal{G}_i.$$

Therefore

$$\sigma(\mathcal{A}_i) = \mathcal{G}_i.$$

Now take

$$A_i \in \mathcal{A}_i,$$

where

$$A_i = \bigcap_{j=1}^{m(i)} A_{i,j}, \quad A_{i,j} \in \mathcal{F}_{i,j}.$$

Since all the $\mathcal{F}_{i,j}$ are independent,

$$\begin{aligned} \mathbb{P} \left(\bigcap_{i=1}^n A_i \right) &= \mathbb{P} \left(\bigcap_{i=1}^n \bigcap_{j=1}^{m(i)} A_{i,j} \right) \\ &= \prod_{i=1}^n \prod_{j=1}^{m(i)} \mathbb{P}(A_{i,j}). \end{aligned}$$

For each i ,

$$A_i = \bigcap_{j=1}^{m(i)} A_{i,j},$$

and hence, again by independence,

$$\mathbb{P}(A_i) = \prod_{j=1}^{m(i)} \mathbb{P}(A_{i,j}).$$

Therefore

$$\mathbb{P} \left(\bigcap_{i=1}^n A_i \right) = \prod_{i=1}^n \mathbb{P}(A_i).$$

Thus

$$\mathcal{A}_1, \dots, \mathcal{A}_n$$

are independent π -systems.

By Theorem ??,

$$\sigma(\mathcal{A}_1), \dots, \sigma(\mathcal{A}_n)$$

are independent.

Since

$$\sigma(\mathcal{A}_i) = \mathcal{G}_i,$$

we conclude that

$$\boxed{\mathcal{G}_1, \dots, \mathcal{G}_n \text{ are independent.}}$$

□

Theorem 3. Functions of Independent Random Variables

Suppose that, for $1 \leq i \leq n$, $1 \leq j \leq m(i)$, the random variables $X_{i,j}$ are independent.

Suppose also that $f_i : \mathbb{R}^{m(i)} \rightarrow \mathbb{R}$ are measurable functions.

Then the random variables $f_i(X_{i,1}, \dots, X_{i,m(i)}), \quad 1 \leq i \leq n$, are independent.

Proof. For each i, j , let

$$\mathcal{F}_{i,j} = \sigma(X_{i,j})$$

and define

$$\mathcal{G}_i = \sigma\left(\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j}\right).$$

Since the random variables $X_{i,j}$ are independent, the σ -fields

$$\mathcal{F}_{i,j}$$

are independent.

Hence, by Theorem ??,

$$\mathcal{G}_1, \dots, \mathcal{G}_n$$

are independent.

Now fix i .

Since

$$X_{i,j}$$

is $\mathcal{F}_{i,j}$ -measurable and

$$\mathcal{F}_{i,j} \subseteq \mathcal{G}_i,$$

each $X_{i,j}$ is $\mathcal{G}_i$ -measurable.

Therefore

$$(X_{i,1}, \dots, X_{i,m(i)})$$

is $\mathcal{G}_i$ -measurable.

Since f_i is measurable, the composition

$$Y_i = f_i(X_{i,1}, \dots, X_{i,m(i)})$$

is $\mathcal{G}_i$ -measurable.

Consequently,

$$\sigma(Y_i) \subseteq \mathcal{G}_i.$$

It remains to prove that

$$Y_1, \dots, Y_n$$

are independent.

Take arbitrary

$$B_i \in \sigma(Y_i), \quad 1 \leq i \leq n.$$

Since

$$\sigma(Y_i) \subseteq \mathcal{G}_i,$$

we have

$$B_i \in \mathcal{G}_i$$

for every i .

But

$$\mathcal{G}_1, \dots, \mathcal{G}_n$$

are independent. Hence

$$\mathbb{P}\left(\bigcap_{i=1}^n B_i\right) = \prod_{i=1}^n \mathbb{P}(B_i).$$

Since this holds for every choice

$$B_i \in \sigma(Y_i),$$

the σ -fields

$$\sigma(Y_1), \dots, \sigma(Y_n)$$

are independent.

Therefore

$$Y_1, \dots, Y_n$$

are independent.

Recalling that

$$Y_i = f_i(X_{i,1}, \dots, X_{i,m(i)}),$$

we obtain

$$f_1(X_{1,1}, \dots, X_{1,m(1)}), \dots, f_n(X_{n,1}, \dots, X_{n,m(n)}) \text{ are independent.}$$

□

Theorem 4. Product Distribution of Independent Random Variables

Suppose $X_1, \dots, X_n$ are independent random variables and X_i has distribution μ_i . Then the random vector $(X_1, \dots, X_n)$ has distribution $\mu_1 \times \dots \times \mu_n$.

Proof. Let

$$X = (X_1, \dots, X_n).$$

The distribution of X is the probability measure μ on $\mathbb{R}^n$ defined by

$$\mu(B) = \mathbb{P}(X \in B), \quad B \in \mathcal{B}(\mathbb{R}^n).$$

Consider the collection

$$\mathcal{A} = \{A_1 \times \dots \times A_n : A_i \in \mathcal{B}(\mathbb{R})\}.$$

This is a π -system generating $\mathcal{B}(\mathbb{R}^n)$.

For

$$A = A_1 \times \dots \times A_n \in \mathcal{A},$$

we have

$$\mu(A) = \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n).$$

Since

$$X_1, \dots, X_n$$

are independent,

$$\begin{aligned} \mu(A) &= \prod_{i=1}^n \mathbb{P}(X_i \in A_i) \\ &= \prod_{i=1}^n \mu_i(A_i). \end{aligned}$$

But the product measure satisfies

$$(\mu_1 \times \cdots \times \mu_n)(A_1 \times \cdots \times A_n) = \prod_{i=1}^n \mu_i(A_i).$$

Therefore

$$\mu(A) = (\mu_1 \times \cdots \times \mu_n)(A)$$

for every

$$A \in \mathcal{A}.$$

Since $\mathcal{A}$ is a π -system generating $\mathcal{B}(\mathbb{R}^n)$, the uniqueness theorem for probability measures implies

$$\mu = \mu_1 \times \cdots \times \mu_n.$$

Hence

$$\mathcal{L}(X_1, \dots, X_n) = \mu_1 \times \cdots \times \mu_n.$$

□

11.1 Sum and Product of Random Variables

11.1.1 Sums of Independent Random Variables

Theorem 5. Distribution Function of the Sum

Let X and Y be independent random variables with distribution functions

$$F(x) = \mathbb{P}(X \leq x), \quad G(y) = \mathbb{P}(Y \leq y).$$

Then

$$\mathbb{P}(X + Y \leq z) = \int_{\mathbb{R}} F(z - y) dG(y).$$

Proof. Define

$$h(x, y) = \mathbf{1}_{\{x+y \leq z\}}.$$

Let μ and ν denote the probability distributions of X and Y , respectively. Thus

$$\mu(A) = \mathbb{P}(X \in A), \quad \nu(B) = \mathbb{P}(Y \in B).$$

Since X and Y are independent, their joint distribution is

$$\mu \times \nu.$$

Therefore

$$\begin{aligned} \mathbb{P}(X + Y \leq z) &= \mathbb{P}((X, Y) \in \{(x, y) : x + y \leq z\}) \\ &= \iint \mathbf{1}_{\{x+y \leq z\}} \mu(dx) \nu(dy). \end{aligned}$$

For fixed y ,

$$h(x, y) = \mathbf{1}_{(-\infty, z-y]}(x).$$

Hence

$$\begin{aligned} \int h(x, y) \mu(dx) &= \int \mathbf{1}_{(-\infty, z-y]}(x) \mu(dx) \\ &= \mu((-\infty, z-y]) \\ &= \mathbb{P}(X \leq z-y) \\ &= F(z-y). \end{aligned}$$

Since $h \geq 0$, Tonelli's theorem allows us to integrate first with respect to x . Thus

$$\begin{aligned} \mathbb{P}(X + Y \leq z) &= \iint \mathbf{1}_{\{x+y \leq z\}} \mu(dx) \nu(dy) \\ &= \int \left[\int \mathbf{1}_{\{x+y \leq z\}} \mu(dx) \right] \nu(dy) \\ &= \int F(z-y) \nu(dy). \end{aligned}$$

Since G is the distribution function of Y , integration with respect to ν is conventionally written as integration with respect to $dG(y)$. Therefore

$$\boxed{\mathbb{P}(X + Y \leq z) = \int_{\mathbb{R}} F(z-y) dG(y).}$$

□

Theorem 6. Density of the Sum

Suppose that X has density f and Y has distribution function G , and that X and Y are independent.

Then $X + Y$ has density

$$h(x) = \int_{\mathbb{R}} f(x-y) dG(y).$$

If Y also has a density g , then

$$h(x) = \int_{\mathbb{R}} f(x-y)g(y) dy.$$

Proof. For any $z \in \mathbb{R}$, by the preceding theorem,

$$\mathbb{P}(X + Y \leq z) = \int_{\mathbb{R}} F(z-y) dG(y).$$

Since X has density f ,

$$F(z-y) = \int_{-\infty}^{z-y} f(u) du.$$

Making the change of variable

$$x = u + y,$$

so that

$$u = x - y, \quad du = dx,$$

we obtain

$$F(z - y) = \int_{-\infty}^z f(x - y) dx.$$

Consequently,

$$\mathbb{P}(X + Y \leq z) = \int_{\mathbb{R}} \left[\int_{-\infty}^z f(x - y) dx \right] dG(y).$$

Since $f \geq 0$, Tonelli's theorem allows us to interchange the order of integration:

$$\mathbb{P}(X + Y \leq z) = \int_{-\infty}^z \left[\int_{\mathbb{R}} f(x - y) dG(y) \right] dx.$$

Define

$$h(x) = \int_{\mathbb{R}} f(x - y) dG(y).$$

Then

$$\mathbb{P}(X + Y \leq z) = \int_{-\infty}^z h(x) dx.$$

Thus h is a density of $X + Y$, and hence

$$h(x) = \int_{\mathbb{R}} f(x - y) dG(y).$$

Finally, if Y has density g , then

$$dG(y) = g(y) dy.$$

Therefore

$$h(x) = \int_{\mathbb{R}} f(x - y) g(y) dy.$$

This is the convolution formula for the density of the sum of two independent random variables.

□

End of Lecture

Probability Theory

Lecture 12

Introduction to Probability

Date: Wednesday, 18 September 2026

Instructor: Dr. Nishant Chandgotia

Observation

Suppose that $\mu = f \, dm$, $\nu = g \, dm$.

Then the product measure is $\mu \times \nu$, and the convolution of μ and ν has density $f * g$. More precisely,

$$\mu * \nu = (f * g) \, dm,$$

where

$$(f * g)(z) = \int_{\mathbb{R}} f(z - y)g(y) \, dy.$$

A proof of this result can be found in Durrett.

Observation

If the underlying measure is the counting measure, then the convolution takes the simple form

$$(f * g)(n) = \sum_m f(m)g(n - m).$$

Thus, in the discrete case, convolution is given by a sum rather than an integral.

12.1 Distribution Function of a Sum

Theorem 1. Distribution Function of a Sum

Let $X \sim \mu$, $Y \sim \nu$, where X and Y are independent.

If F and G are the distribution functions of μ and ν , respectively, then the distribution function of $X + Y$ is

$$\begin{aligned} \mathbb{P}(X + Y \leq z) &= \mathbb{P}((X, Y) \in \{(x, y) : x + y \leq z\}) \\ &= \int_{\{x+y \leq z\}} d(\mu \times \nu)(x, y). \end{aligned}$$

Equivalently,

$$\mathbb{P}(X + Y \leq z) = \int_{\mathbb{R}} F(z - y) \, dG(y).$$

12.2 Some Questions and Exercises on Sum and Product of Random Variables

Question 1. Density of the Product of Two Independent Random Variables

Let $X \sim \mu$, $Y \sim \nu$, where $\mu = f dm$, $\nu = g dm$.
Assume that X and Y are independent. Compute the density of XY .

Question 2. Sum of Two Independent Poisson Random Variables

Let $X \sim \text{Poi}(\lambda)$, $Y \sim \text{Poi}(\mu)$, and suppose that X and Y are independent.
Determine the distribution of $X + Y$.

Question 3. Sum of Two Independent Binomial Random Variables

Let $X \sim \text{Bin}(m, p)$, $Y \sim \text{Bin}(m, p)$, and suppose that X and Y are independent.
Determine the distribution of $X + Y$.

Example 1. Sum of Two Independent Exponential Random Variables

Suppose $X, Y \sim \text{Exp}(\lambda)$ are independent, where $\lambda > 0$ is the rate parameter.
The densities of X and Y are

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0,$$

and

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0.$$

Let $Z = X + Y$.

By the convolution formula,

$$f_Z(y) = \int_0^y f_X(x) f_Y(y-x) dx.$$

Therefore,

$$\begin{aligned} f_Z(y) &= \int_0^y \lambda e^{-\lambda x} \lambda e^{-\lambda(y-x)} dx \\ &= \lambda^2 e^{-\lambda y} \int_0^y dx \\ &= \lambda^2 y e^{-\lambda y}, \quad y > 0. \end{aligned}$$

Thus $X + Y$ has a Gamma distribution with shape parameter 2 and rate parameter λ :
 $X + Y \sim \Gamma(2, \lambda)$.

Proposition 1. Sum of Independent Exponential Random Variables

Let $X_1, X_2, \dots, X_n$ be independent random variables such that $X_i \sim \text{Exp}(\lambda)$, $\lambda > 0$.

Then

$$X_1 + \cdots + X_n \sim \Gamma(n, \lambda).$$

Its density is

$$f(x) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x}, \quad x > 0.$$

Proof. We prove the result by induction on n .

For $n = 1$,

$$f_{X_1}(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

Since $\text{Gamma}(1, \lambda)$ has density $\frac{\lambda^1}{\Gamma(1)} x^{1-1} e^{-\lambda x} = \lambda e^{-\lambda x}$, the result holds for $n = 1$.

Suppose the result holds for n . Then $S_n = X_1 + \cdots + X_n$ has density

$$f_{S_n}(x) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x}, \quad x > 0.$$

Since X_{n+1} is independent of S_n , $S_{n+1} = S_n + X_{n+1}$ has density given by convolution:

$$\begin{aligned} f_{S_{n+1}}(x) &= \int_0^x f_{S_n}(u) f_{X_{n+1}}(x-u) du \\ &= \int_0^x \frac{\lambda^n}{(n-1)!} u^{n-1} e^{-\lambda u} \lambda e^{-\lambda(x-u)} du \\ &= \frac{\lambda^{n+1} e^{-\lambda x}}{(n-1)!} \int_0^x u^{n-1} du \\ &= \frac{\lambda^{n+1} e^{-\lambda x}}{(n-1)!} \frac{x^n}{n} \\ &= \frac{\lambda^{n+1}}{n!} x^n e^{-\lambda x}. \end{aligned}$$

Thus S_{n+1} has density $f_{S_{n+1}}(x) = \frac{\lambda^{n+1}}{n!} x^n e^{-\lambda x}$, $x > 0$.

Therefore, by induction,

$$X_1 + \cdots + X_n \sim \Gamma(n, \lambda).$$

□

Definition 1. Gamma Distribution

Let $\alpha > 0$, $\lambda > 0$.

A random variable X is said to have a Gamma distribution with shape parameter α and rate parameter λ if its density is

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0,$$

and

$$f_X(x) = 0, \quad x \leq 0.$$

We write

$$X \sim \text{Gamma}(\alpha, \lambda).$$

The Gamma function is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \quad \alpha > 0.$$

For $n \in \mathbb{N}$,

$$\Gamma(n) = (n-1)!.$$

Proposition 2. Sum of Independent Gamma Random Variables

Let

$$X \sim \text{Gamma}(\alpha, \lambda), \quad Y \sim \text{Gamma}(\beta, \lambda),$$

where X and Y are independent and λ is the common rate parameter.

Then

$$X + Y \sim \text{Gamma}(\alpha + \beta, \lambda).$$

Proof. Let

$$Z = X + Y.$$

Since X and Y are independent, the density of Z is given by the convolution of the densities of X and Y :

$$f_Z(z) = \int_{-\infty}^\infty f_X(x) f_Y(z-x) dx.$$

Since $X > 0$ and $Y > 0$, for $z > 0$ we must have $x > 0$ and $z - x > 0$.

Thus

$$0 < x < z.$$

Therefore,

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx.$$

Substituting the Gamma densities,

$$\begin{aligned} f_Z(z) &= \int_0^z \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \frac{\lambda^\beta}{\Gamma(\beta)} (z-x)^{\beta-1} e^{-\lambda(z-x)} dx \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^z x^{\alpha-1} (z-x)^{\beta-1} e^{-\lambda x} e^{-\lambda(z-x)} dx. \end{aligned}$$

But

$$e^{-\lambda x} e^{-\lambda(z-x)} = e^{-\lambda z}.$$

Hence

$$f_Z(z) = \frac{\lambda^{\alpha+\beta} e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^z x^{\alpha-1} (z-x)^{\beta-1} dx.$$

We now evaluate the integral

$$I = \int_0^z x^{\alpha-1} (z-x)^{\beta-1} dx.$$

Make the substitution $u = \frac{x}{z}$.

Then $x = zu$, $dx = z du$, and $z - x = z(1 - u)$.

When $x = 0$, $u = 0$, and when $x = z$, $u = 1$. Therefore,

$$\begin{aligned} I &= \int_0^1 (zu)^{\alpha-1} (z(1-u))^{\beta-1} z du \\ &= z^{\alpha+\beta-1} \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du. \end{aligned}$$

Recall the Beta function:

$$B(\alpha, \beta) = \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du.$$

The Beta–Gamma identity gives

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Consequently,

$$I = z^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Substituting this into the expression for $f_Z(z)$,

$$\begin{aligned} f_Z(z) &= \frac{\lambda^{\alpha+\beta} e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)} z^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)} \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha + \beta)} z^{\alpha+\beta-1} e^{-\lambda z}, \quad z > 0. \end{aligned}$$

Also,

$$f_Z(z) = 0, \quad z \leq 0.$$

But this is precisely the density of

$$\text{Gamma}(\alpha + \beta, \lambda).$$

Therefore,

$$\boxed{X + Y \sim \text{Gamma}(\alpha + \beta, \lambda).}$$

□

Recall the Carathéodory Extension Theorem.

End of Lecture

Probability Theory

Lecture 13

Introduction to Probability

Date: Monday, 21 September 2026

Instructor: Dr. Nishant Chandgotia

13.1 Kolmogorov Extension Theorem

Theorem 1. Kolmogorov's Extension Theorem

For every $n \geq 1$, let μ_n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$.

Suppose that the family $\{\mu_n : n \geq 1\}$ is consistent in the following sense: for every $n \geq 1$ and every Borel set $B \in \mathcal{B}(\mathbb{R}^n)$, we have

$$\mu_{n+1}(B \times \mathbb{R}) = \mu_n(B).$$

Then there exists a unique probability measure $\mathbb{P}$ on $(\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}})$ such that

$$\mathbb{P}(B \times \mathbb{R}^{\mathbb{N} \setminus \{1, \dots, n\}}) = \mu_n(B)$$

for every

$$B \in \mathcal{B}(\mathbb{R}^n).$$

Equivalently, for every finite-dimensional rectangle

$$B = (a_1, b_1] \times \cdots \times (a_n, b_n],$$

we have

$$\mathbb{P}\{\omega \in \mathbb{R}^{\mathbb{N}} : \omega_i \in (a_i, b_i], 1 \leq i \leq n\} = \mu_n(B).$$

Remark

The spaces $\mathbb{R}$ and $\mathbb{N}$ in the above formulation cannot simply be replaced by arbitrary spaces without additional assumptions.

More generally, the theorem can be formulated for suitable measurable spaces. In particular, the relevant generalization is closely related to standard Borel spaces.

A proof of the theorem can be found in Durrett, Theorem 2.1.21.

13.1.1 Proof of the Kolmogorov Extension Theorem

Proof. We divide the proof into several steps.

Step 1: The elementary rectangles.

Let $\mathcal{S}$ be the collection of subsets of $\mathbb{R}^{\mathbb{N}}$ of the form

$$A = \left\{ \omega \in \mathbb{R}^{\mathbb{N}} : \omega_i \in (a_i, b_i], \quad 1 \leq i \leq n \right\},$$

where

$$-\infty \leq a_i < b_i \leq \infty.$$

Equivalently,

$$A = (a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R} \times \mathbb{R} \times \cdots.$$

The collection $\mathcal{S}$ is a semialgebra of subsets of $\mathbb{R}^{\mathbb{N}}$.

Indeed, the intersection of two such sets is again of the same form, after taking the larger of the two dimensions.

Moreover, the difference of two such rectangles can be written as a finite disjoint union of rectangles of the same type.

Step 2: Definition of $\mathbb{P}$ on $\mathcal{S}$.

For

$$A = \{ \omega : \omega_i \in (a_i, b_i], \quad 1 \leq i \leq n \},$$

define

$$\boxed{\mathbb{P}(A) = \mu_n((a_1, b_1] \times \cdots \times (a_n, b_n])}. \quad (1)$$

We first show that this definition is well-defined.

Suppose that the same set is regarded as a subset determined by the first m coordinates, where $m > n$. Then it is represented by

$$(a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R}^{m-n}.$$

By consistency,

$$\mu_m((a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R}^{m-n}) = \mu_n((a_1, b_1] \times \cdots \times (a_n, b_n]).$$

Indeed, this follows by applying

$$\mu_{k+1}(B \times \mathbb{R}) = \mu_k(B)$$

successively for

$$k = n, n+1, \dots, m-1.$$

Thus the value in (1) does not depend on the number of coordinates used to describe the set.

Step 3: Finite additivity.

Suppose

$$A = A_1 \sqcup \cdots \sqcup A_k,$$

where

$$A, A_1, \dots, A_k \in \mathcal{S}.$$

There exists an integer N such that all these sets depend only on the first N coordinates.

Hence there are rectangles

$$R, R_1, \dots, R_k \subseteq \mathbb{R}^N$$

such that

$$A = R \times \mathbb{R}^{N \setminus \{1, \dots, N\}},$$

and similarly

$$A_j = R_j \times \mathbb{R}^{N \setminus \{1, \dots, N\}}.$$

Moreover,

$$R = R_1 \sqcup \dots \sqcup R_k.$$

Since μ_N is a probability measure,

$$\mu_N(R) = \sum_{j=1}^k \mu_N(R_j).$$

Therefore,

$$\mathbb{P}(A) = \sum_{j=1}^k \mathbb{P}(A_j).$$

Thus $\mathbb{P}$ is finitely additive on $\mathcal{S}$.

Step 4: Continuity lemma.

Let $\mathcal{A}$ be the algebra generated by $\mathcal{S}$.

We claim that if

$$B_n \in \mathcal{A}, \quad B_n \downarrow \emptyset,$$

then

$$\boxed{\mathbb{P}(B_n) \downarrow 0.} \tag{2}$$

Suppose, to the contrary, that

$$\mathbb{P}(B_n) \downarrow \delta$$

for some

$$\delta > 0.$$

Since each B_n belongs to $\mathcal{A}$, it can be written as a finite disjoint union of finite-dimensional rectangles. Thus, after representing all rectangles using the first n coordinates, we may write

$$B_n = \bigcup_{k=1}^{K_n} \left\{ \omega : \omega_i \in (a_i^k, b_i^k], \quad 1 \leq i \leq n \right\}. \tag{3}$$

For every n , choose a finite union $C_n \subseteq B_n$ of bounded closed rectangles such that

$$\boxed{\mathbb{P}(B_n \setminus C_n) \leq \frac{\delta}{2^{n+1}}.} \tag{4}$$

Such an approximation is possible because μ_n is a probability measure on $\mathbb{R}^n$.

Now define

$$D_n = \bigcap_{m=1}^n C_m.$$

Since

$$C_m \subseteq B_m$$

and

$$B_n \subseteq B_m \quad (m \leq n),$$

we have

$$D_n \subseteq B_n.$$

Furthermore,

$$B_n \setminus D_n \subseteq \bigcup_{m=1}^n (B_m \setminus C_m).$$

Therefore, by subadditivity,

$$\begin{aligned} \mathbb{P}(B_n \setminus D_n) &\leq \sum_{m=1}^n \mathbb{P}(B_m \setminus C_m) \\ &\leq \sum_{m=1}^n \frac{\delta}{2^{m+1}} \\ &< \frac{\delta}{2}. \end{aligned}$$

Since

$$\mathbb{P}(B_n) \geq \delta,$$

we obtain

$$\begin{aligned} \mathbb{P}(D_n) &= \mathbb{P}(B_n) - \mathbb{P}(B_n \setminus D_n) \\ &\geq \delta - \frac{\delta}{2} \\ &= \frac{\delta}{2}. \end{aligned}$$

Thus

$$\boxed{\mathbb{P}(D_n) \geq \frac{\delta}{2} > 0.} \tag{5}$$

In particular,

$$D_n \neq \emptyset$$

for every n .

Step 5: Finite-dimensional compactness argument.

Since D_n depends only on the first n coordinates, there exists a set

$$D_n^* \subseteq \mathbb{R}^n$$

such that

$$D_n = D_n^* \times \mathbb{R} \times \mathbb{R} \times \cdots.$$

The set D_n^* is a finite union of bounded closed rectangles and hence is compact.

For each n , choose

$$\omega^{(n)} \in D_n.$$

Write

$$\omega^{(n)} = (\omega_1^{(n)}, \omega_2^{(n)}, \dots).$$

Since

$$D_n \subseteq D_1,$$

we have

$$\omega_1^{(n)} \in D_1^*.$$

By compactness of D_1^* , there exists a subsequence such that

$$\omega_1^{(n)} \longrightarrow \theta_1$$

for some

$$\theta_1 \in D_1^*.$$

Next, since

$$D_n \subseteq D_2 \quad (n \geq 2),$$

we have

$$(\omega_1^{(n)}, \omega_2^{(n)}) \in D_2^*.$$

Using compactness of D_2^* , we can extract a further subsequence such that

$$\omega_2^{(n)} \longrightarrow \theta_2.$$

The first coordinate still converges to θ_1 because we have only passed to a subsequence.

Continuing inductively, we obtain

$$\theta_1, \theta_2, \dots$$

such that, for every k ,

$$\boxed{(\theta_1, \dots, \theta_k) \in D_k^*}. \tag{6}$$

This is the diagonal-subsequence argument.

Step 6: Contradiction.

Define

$$\theta = (\theta_1, \theta_2, \dots) \in \mathbb{R}^{\mathbb{N}}.$$

By (6),

$$(\theta_1, \dots, \theta_k) \in D_k^*,$$

and therefore

$$\theta \in D_k$$

for every k .

Consequently,

$$\theta \in \bigcap_{k=1}^{\infty} D_k.$$

Since

$$D_k \subseteq B_k,$$

we obtain

$$\theta \in \bigcap_{k=1}^{\infty} B_k.$$

Hence

$$\bigcap_{k=1}^{\infty} B_k \neq \emptyset,$$

which contradicts

$$B_k \downarrow \emptyset.$$

Therefore

$$\mathbb{P}(B_n) \downarrow 0.$$

This proves the continuity property (2).

Step 7: Countable additivity.

Suppose

$$A = \bigsqcup_{i=1}^{\infty} A_i, \quad A, A_i \in \mathcal{S}.$$

Define

$$B_n = A \setminus \bigcup_{i=1}^n A_i.$$

Then

$$B_n \downarrow \emptyset.$$

Also,

$$A = A_1 \sqcup \cdots \sqcup A_n \sqcup B_n.$$

By finite additivity,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A_i) + \mathbb{P}(B_n).$$

By the continuity property,

$$\mathbb{P}(B_n) \longrightarrow 0.$$

Therefore,

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Thus $\mathbb{P}$ is countably additive.

Step 8: Extension to the product σ -field.

We have

$$\mathbb{P}(\mathbb{R}^{\mathbb{N}}) = \mu_1(\mathbb{R}) = 1.$$

Thus $\mathbb{P}$ is a finite measure on the algebra generated by $\mathcal{S}$.

By the Carathéodory Extension Theorem, $\mathbb{P}$ extends to a measure on

$$\sigma(\mathcal{S}).$$

But

$$\sigma(\mathcal{S}) = \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}}.$$

Therefore there exists a probability measure

$$\mathbb{P}$$

on

$$(\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}})$$

having the prescribed finite-dimensional distributions.

Thus existence is proved.

Step 9: Uniqueness.

Suppose that $\mathbb{P}$ and $\mathbb{Q}$ are two probability measures on

$$(\mathbb{R}^{\mathbb{N}}, \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}})$$

having the prescribed finite-dimensional distributions.

Then

$$\mathbb{P}(A) = \mathbb{Q}(A)$$

for every finite-dimensional rectangle

$$A \in \mathcal{S}.$$

Define

$$\mathcal{L} = \{A : \mathbb{P}(A) = \mathbb{Q}(A)\}.$$

We show that $\mathcal{L}$ is a λ -system.

First,

$$\mathbb{P}(\Omega) = \mathbb{Q}(\Omega) = 1,$$

so

$$\Omega \in \mathcal{L}.$$

Suppose

$$A, B \in \mathcal{L}, \quad A \subseteq B.$$

Then

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$$

and

$$\mathbb{Q}(B \setminus A) = \mathbb{Q}(B) - \mathbb{Q}(A).$$

Since

$$\mathbb{P}(A) = \mathbb{Q}(A)$$

and

$$\mathbb{P}(B) = \mathbb{Q}(B),$$

we obtain

$$\mathbb{P}(B \setminus A) = \mathbb{Q}(B \setminus A).$$

Hence

$$B \setminus A \in \mathcal{L}.$$

Finally, suppose

$$A_n \uparrow A$$

with

$$A_n \in \mathcal{L}.$$

By continuity from below,

$$\begin{aligned} \mathbb{P}(A) &= \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \\ &= \lim_{n \rightarrow \infty} \mathbb{Q}(A_n) \\ &= \mathbb{Q}(A). \end{aligned}$$

Thus

$$A \in \mathcal{L}.$$

Therefore $\mathcal{L}$ is a λ -system.

Since $\mathcal{S}$ is a π -system and

$$\mathcal{S} \subseteq \mathcal{L},$$

the π - λ theorem gives

$$\sigma(\mathcal{S}) \subseteq \mathcal{L}.$$

But

$$\sigma(\mathcal{S}) = \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}}.$$

Hence

$$\mathbb{P}(A) = \mathbb{Q}(A)$$

for every

$$A \in \mathcal{B}(\mathbb{R})^{\otimes \mathbb{N}}.$$

Therefore

$$\mathbb{P} = \mathbb{Q}.$$

Thus the extension is unique.

□

13.1.2 Remarks on the General Formulation

Remark

The space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ may be replaced by a measurable space that is Borel isomorphic to it. The general theory of probability measures on metric spaces provides a broader setting for such extension results.

See, for example, the discussion of probability measures on metric spaces in Parthasarathy.

13.2 Weak and Strong Laws of Large Numbers

13.2.1 Introduction

We now move to the main limit theorems of probability theory, beginning with the Weak Law of Large Numbers and the Strong Law of Large Numbers.

Let

$$X_1, X_2, \dots$$

be random variables and define the partial sums

$$S_n = X_1 + \dots + X_n.$$

The quantities

$$\frac{S_n}{n}$$

represent the empirical average of the random variables.

Observation

Under suitable assumptions, the Weak Law of Large Numbers gives convergence of the empirical average in probability, while the Strong Law of Large Numbers gives almost sure convergence.

In the identically distributed case, the limiting quantity is typically $\mathbb{E}(X_1)$.

13.2.2 Example on the Product Space

Example 1. A Sequence of Random Variables on a Product Space

Consider the probability space

$$([0, 1]^{\mathbb{N}}, \mathcal{B}([0, 1])^{\otimes \mathbb{N}}, \mathbb{P}).$$

For

$$\omega = (\omega_1, \omega_2, \dots) \in [0, 1]^{\mathbb{N}},$$

define

$$X_i(\omega) = \begin{cases} H, & \omega_i < \omega_1, \\ T, & \omega_i \geq \omega_1. \end{cases}$$

Thus each X_i is a random variable defined on the product probability space.

13.2.3 Stationary Sequences

Definition 1. Stationary Sequence of Random Variables

A sequence of random variables $X_1, X_2, \dots$ is called *stationary* if, for every $n \geq 1$ and $k \geq 0$, we have

$$(X_1, X_2, \dots, X_n) \stackrel{d}{=} (X_{1+k}, X_{2+k}, \dots, X_{n+k}).$$

Here $\stackrel{d}{=}$ means equality in distribution.

13.2.4 Ergodic Theorem

Observation

Suppose that $X_1, X_2, \dots$ is a stationary sequence satisfying $\mathbb{E}(|X_1|) < \infty$.
Under the assumptions of the ergodic theorem, the empirical averages

$$\frac{S_n}{n}, \quad \text{where } S_n = X_1 + \dots + X_n,$$

converge almost surely to a limiting random variable.

The theorem also identifies the limiting random variable in terms of the invariant σ -field.

Professor's Note

The Kolmogorov Extension Theorem is the basic tool for constructing probability measures on infinite product spaces from consistent finite-dimensional distributions.

The next part of the course develops the Weak and Strong Laws of Large Numbers and then moves toward stationary processes and the Ergodic Theorem.

End of Lecture

13.3 Quiz, 23 September 2026

Quiz 6

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function.

1. Let $S_n \sim \text{Bin}(n, p)$. For all $p \in [0, 1]$, define $f_n(p) = \mathbb{E} \left[f \left(\frac{S_n}{n} \right) \right]$. What are $f_n(0)$ and $f_n(1)$?

2. Prove that $f_n(p)$ is a polynomial.

3. Prove that

$$\mathbb{P} \left(\left| \frac{S_n}{n} - p \right| > \delta \right) \leq 2 \exp \left(-2n\delta^2 \right).$$

4. Prove that $f_n \rightarrow f$ uniformly.

Probability Theory

Lecture 14

Introduction to Probability

Date: Friday, 25 September 2026

Instructor: Dr. Nishant Chandgotia

14.1 Weak Law of Large Numbers

14.1.1 Weak Law Under a Uniform Variance Bound

Suppose that $X_1, X_2, \dots, X_n, \dots$ are uncorrelated random variables satisfying $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) \leq c$ for some constant $c > 0$ and all i .

Let

$$S_n = X_1 + \dots + X_n.$$

Then $\mathbb{E}(S_n) = n\mu$ and, since the random variables are uncorrelated,

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) \leq nc.$$

Consequently,

$$\begin{aligned} \mathbb{E} \left[\left(\frac{S_n}{n} - \mu \right)^2 \right] &= \frac{1}{n^2} \text{Var}(S_n) \\ &\leq \frac{c}{n} \rightarrow 0. \end{aligned}$$

Hence

$$\boxed{\frac{S_n}{n} \rightarrow \mu \text{ in } L^2.}$$

Since convergence in L^2 implies convergence in probability, we also have

$$\boxed{\frac{S_n}{n} \xrightarrow{P} \mu.}$$

Observation

The uniform variance bound $\text{Var}(X_i) \leq c$ is a strong assumption. Our goal is to obtain a Weak Law of Large Numbers under weaker conditions.

14.2 Truncation

14.2.1 Truncated Random Variables

A standard way to remove the strong variance assumption is to truncate the random variables.

For a sequence of random variables $X_1, X_2, \dots$, define

$$\bar{X}_i = X_i \mathbf{1}_{\{|X_i| \leq i\}}.$$

More generally, for a truncation level $b_n > 0$, define

$$\bar{X}_{n,k} = X_{n,k} \mathbf{1}_{\{|X_{n,k}| \leq b_n\}}.$$

The idea is that the truncated variables have better controlled second moments, while the probability that truncation actually changes the variables can be made small.

We are interested in the relationship between $\frac{1}{n} \sum_{i=1}^n \bar{X}_i$ and $\frac{1}{n} \sum_{i=1}^n X_i$.

This leads naturally to the Weak Law for triangular arrays.

14.3 Weak Law for Triangular Arrays

14.3.1 Statement

Theorem 1. Weak Law for Triangular Arrays

For each $n \geq 1$, let $X_{n,k}$, $1 \leq k \leq n$, be independent random variables.

Let $b_n > 0$, $b_n \rightarrow \infty$, and define the truncated variables

$$\bar{X}_{n,k} = X_{n,k} \mathbf{1}_{\{|X_{n,k}| \leq b_n\}}.$$

Suppose that, as $n \rightarrow \infty$,

$$\sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n) \rightarrow 0, \tag{i}$$

$$b_n^{-2} \sum_{k=1}^n \mathbb{E} [\bar{X}_{n,k}^2] \rightarrow 0. \tag{ii}$$

Let

$$S_n = X_{n,1} + \dots + X_{n,n}$$

and define

$$a_n = \sum_{k=1}^n \mathbb{E} [\bar{X}_{n,k}].$$

Then

$$\boxed{\frac{S_n - a_n}{b_n} \xrightarrow{P} 0.}$$

14.3.2 Proof

Proof. Define

$$\bar{S}_n = \bar{X}_{n,1} + \cdots + \bar{X}_{n,n}.$$

We shall show that, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \rightarrow 0.$$

Since

$$S_n - a_n = (S_n - \bar{S}_n) + (\bar{S}_n - a_n),$$

we have

$$\left\{\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right\} \subseteq \{S_n \neq \bar{S}_n\} \cup \left\{\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right\}.$$

Hence, by the union bound,

$$\mathbb{P}\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \leq \mathbb{P}(S_n \neq \bar{S}_n) + \mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right). \quad (1)$$

Step 1: Probability that truncation changes the sum.

If

$$|X_{n,k}| \leq b_n,$$

then

$$\bar{X}_{n,k} = X_{n,k} \mathbf{1}_{\{|X_{n,k}| \leq b_n\}} = X_{n,k}.$$

Therefore,

$$\{S_n \neq \bar{S}_n\} \subseteq \bigcup_{k=1}^n \{\bar{X}_{n,k} \neq X_{n,k}\}.$$

Moreover,

$$\{\bar{X}_{n,k} \neq X_{n,k}\} \subseteq \{|X_{n,k}| > b_n\}.$$

Hence

$$\{S_n \neq \bar{S}_n\} \subseteq \bigcup_{k=1}^n \{|X_{n,k}| > b_n\}.$$

Consequently,

$$\begin{aligned} \mathbb{P}(S_n \neq \bar{S}_n) &\leq \mathbb{P}\left(\bigcup_{k=1}^n \{|X_{n,k}| > b_n\}\right) \\ &\leq \sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n). \end{aligned}$$

By assumption (i),

$$\boxed{\mathbb{P}(S_n \neq \bar{S}_n) \rightarrow 0.} \quad (2)$$

Step 2: Control of the truncated sum.

By linearity of expectation,

$$\begin{aligned}\mathbb{E}(\bar{S}_n) &= \mathbb{E}\left[\sum_{k=1}^n \bar{X}_{n,k}\right] \\ &= \sum_{k=1}^n \mathbb{E}(\bar{X}_{n,k}) \\ &= a_n.\end{aligned}$$

Thus

$$\bar{S}_n - a_n = \bar{S}_n - \mathbb{E}(\bar{S}_n).$$

Therefore, by Chebyshev's inequality,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right) &= \mathbb{P}\left(|\bar{S}_n - \mathbb{E}(\bar{S}_n)| > \varepsilon b_n\right) \\ &\leq \frac{\text{Var}(\bar{S}_n)}{\varepsilon^2 b_n^2}.\end{aligned}\tag{3}$$

Since the $X_{n,k}$ are independent, the truncated random variables $\bar{X}_{n,k}$ are also independent. Hence

$$\text{Var}(\bar{S}_n) = \sum_{k=1}^n \text{Var}(\bar{X}_{n,k}).$$

For any random variable Y ,

$$\text{Var}(Y) = \mathbb{E}(Y^2) - (\mathbb{E}Y)^2 \leq \mathbb{E}(Y^2).$$

Therefore,

$$\text{Var}(\bar{X}_{n,k}) \leq \mathbb{E}[\bar{X}_{n,k}^2].$$

Substituting this into (3), we obtain

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right) &\leq \frac{1}{\varepsilon^2 b_n^2} \sum_{k=1}^n \text{Var}(\bar{X}_{n,k}) \\ &\leq \frac{1}{\varepsilon^2 b_n^2} \sum_{k=1}^n \mathbb{E}[\bar{X}_{n,k}^2] \\ &= \frac{1}{\varepsilon^2} \left[b_n^{-2} \sum_{k=1}^n \mathbb{E}[\bar{X}_{n,k}^2] \right].\end{aligned}$$

By assumption (ii),

$$\boxed{\mathbb{P}\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right) \longrightarrow 0.}\tag{4}$$

Step 3: Conclusion.

Combining (1), (2), and (4), we obtain

$$\mathbb{P}\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \longrightarrow 0.$$

Since this holds for every $\varepsilon > 0$,

$$\frac{S_n - a_n}{b_n} \xrightarrow{P} 0.$$

Hence the proof is complete. □

14.4 A Weak Law Under a Tail Condition

14.4.1 Truncated Weak Law of Large Numbers

Theorem 2. Weak Law of Large Numbers Under a Tail Condition

Let $X_1, X_2, \dots$ be i.i.d. random variables satisfying

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0 \quad \text{as } x \rightarrow \infty.$$

Let $S_n = X_1 + \dots + X_n$ and define $\mu_n = \mathbb{E} \left[X_1 \mathbf{1}_{\{|X_1| \leq n\}} \right]$.

Then

$$\frac{S_n}{n} - \mu_n \xrightarrow{P} 0.$$

14.4.2 Proof

Proof. We apply Theorem Weak Law For Triangular Array to the triangular array

$$X_{n,k} = X_k, \quad 1 \leq k \leq n,$$

with

$$b_n = n.$$

Since

$$X_1, X_2, \dots$$

are i.i.d., the variables

$$X_{n,1}, \dots, X_{n,n}$$

are independent for every n .

Verification of condition (i).

We have

$$\begin{aligned} \sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n) &= \sum_{k=1}^n \mathbb{P}(|X_k| > n) \\ &= n\mathbb{P}(|X_1| > n). \end{aligned}$$

By assumption,

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0.$$

Taking $x = n$, we obtain

$$n\mathbb{P}(|X_1| > n) \longrightarrow 0.$$

Hence

$$\boxed{\sum_{k=1}^n \mathbb{P}(|X_{n,k}| > b_n) \longrightarrow 0.} \quad (1)$$

Verification of condition (ii).

We have

$$\begin{aligned} b_n^{-2} \sum_{k=1}^n \mathbb{E} \left[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| \leq b_n\}} \right] &= \frac{1}{n^2} \sum_{k=1}^n \mathbb{E} \left[X_k^2 \mathbf{1}_{\{|X_k| \leq n\}} \right] \\ &= \frac{1}{n} \mathbb{E} \left[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}} \right]. \end{aligned} \quad (2)$$

It remains to show that

$$\frac{1}{n} \mathbb{E} \left[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}} \right] \longrightarrow 0. \quad (3)$$

Put

$$Y = |X_1|.$$

Then

$$\mathbb{E} \left[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}} \right] = \mathbb{E} \left[Y^2 \mathbf{1}_{\{Y \leq n\}} \right].$$

Since

$$Y^2 \mathbf{1}_{\{Y \leq n\}} \leq Y^2 \wedge n^2,$$

we have

$$\mathbb{E} \left[Y^2 \mathbf{1}_{\{Y \leq n\}} \right] \leq \mathbb{E}(Y^2 \wedge n^2).$$

Using the tail-integral formula,

$$\mathbb{E}(Y^2 \wedge n^2) = \int_0^{n^2} \mathbb{P}(Y^2 > t) dt.$$

Making the substitution

$$t = x^2, \quad dt = 2x dx,$$

gives

$$\mathbb{E}(Y^2 \wedge n^2) = 2 \int_0^n x \mathbb{P}(Y > x) dx.$$

Consequently,

$$\frac{1}{n} \mathbb{E} \left[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}} \right] \leq \frac{2}{n} \int_0^n x \mathbb{P}(|X_1| > x) dx. \quad (4)$$

By assumption,

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0.$$

Let $\varepsilon > 0$. There exists $M > 0$ such that

$$x\mathbb{P}(|X_1| > x) < \varepsilon \quad \text{for all } x \geq M.$$

For $n > M$,

$$\begin{aligned} \frac{1}{n} \int_0^n x\mathbb{P}(|X_1| > x) dx &= \frac{1}{n} \int_0^M x\mathbb{P}(|X_1| > x) dx \\ &\quad + \frac{1}{n} \int_M^n x\mathbb{P}(|X_1| > x) dx \\ &\leq \frac{1}{n} \int_0^M x\mathbb{P}(|X_1| > x) dx \\ &\quad + \frac{\varepsilon(n-M)}{n}. \end{aligned}$$

As $n \rightarrow \infty$, the first term tends to 0, while

$$\frac{\varepsilon(n-M)}{n} \leq \varepsilon.$$

Hence

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \int_0^n x\mathbb{P}(|X_1| > x) dx \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\frac{1}{n} \int_0^n x\mathbb{P}(|X_1| > x) dx \rightarrow 0.$$

It follows from (4) that

$$\frac{1}{n} \mathbb{E} \left[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}} \right] \rightarrow 0.$$

Thus condition (ii) is satisfied.

Therefore, by Theorem ??,

$$\frac{S_n - a_n}{n} \xrightarrow{P} 0,$$

where

$$a_n = \sum_{k=1}^n \mathbb{E} \left[X_k \mathbf{1}_{\{|X_k| \leq n\}} \right].$$

Since the X_k are identically distributed,

$$\begin{aligned} a_n &= n \mathbb{E} \left[X_1 \mathbf{1}_{\{|X_1| \leq n\}} \right] \\ &= n\mu_n. \end{aligned}$$

Consequently,

$$\frac{S_n - a_n}{n} = \frac{S_n - n\mu_n}{n} = \frac{S_n}{n} - \mu_n.$$

Therefore,

$$\boxed{\frac{S_n}{n} - \mu_n \xrightarrow{P} 0.}$$

The proof is complete. □

Remark

The condition

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0$$

is the tail condition used above to obtain the weak law with the truncated mean μ_n .

14.5 Tail Conditions and Moment Conditions

14.5.1 A Tail Condition Implies Lower Moments

Theorem 3. Tail Condition and Fractional Moments

Let X be a random variable.

If

$$x\mathbb{P}(|X| > x) \longrightarrow 0 \quad \text{as } x \rightarrow \infty,$$

then

$$X \in L^{1-\varepsilon} \quad \text{for every } 0 < \varepsilon < 1.$$

Proof. Suppose that

$$x\mathbb{P}(|X| > x) \longrightarrow 0.$$

We prove that

$$X \in L^p \quad \text{for every } 0 < p < 1.$$

Using the tail-integral formula, for $p > 0$,

$$\mathbb{E}|X|^p = p \int_0^\infty x^{p-1} \mathbb{P}(|X| > x) dx.$$

Since

$$x\mathbb{P}(|X| > x) \longrightarrow 0,$$

there exists $M > 0$ such that

$$x\mathbb{P}(|X| > x) \leq 1 \quad \text{for all } x \geq M.$$

Therefore,

$$\mathbb{P}(|X| > x) \leq \frac{1}{x}, \quad x \geq M.$$

Split the integral at M :

$$\begin{aligned} \mathbb{E}|X|^p &= p \int_0^M x^{p-1} \mathbb{P}(|X| > x) dx \\ &\quad + p \int_M^\infty x^{p-1} \mathbb{P}(|X| > x) dx. \end{aligned}$$

For the first integral,

$$\begin{aligned} p \int_0^M x^{p-1} \mathbb{P}(|X| > x) dx &\leq p \int_0^M x^{p-1} dx \\ &= M^p < \infty. \end{aligned}$$

For the second integral,

$$\begin{aligned} p \int_M^\infty x^{p-1} \mathbb{P}(|X| > x) dx &\leq p \int_M^\infty x^{p-2} dx \\ &= \frac{p}{1-p} M^{p-1} \\ &< \infty, \end{aligned}$$

because

$$0 < p < 1.$$

Therefore,

$$\mathbb{E}|X|^p < \infty.$$

Hence

$$X \in L^p \quad \text{for every } 0 < p < 1.$$

Taking

$$p = 1 - \varepsilon,$$

we obtain

$$X \in L^{1-\varepsilon} \quad \text{for every } 0 < \varepsilon < 1.$$

□

14.5.2 Integrability Implies the Tail Condition

Theorem 4. Integrability Implies Vanishing Tail

Let X be a random variable.

If

$$X \in L^1,$$

then

$$x\mathbb{P}(|X| > x) \longrightarrow 0 \quad \text{as } x \rightarrow \infty.$$

Proof. Suppose that

$$X \in L^1.$$

Then

$$\mathbb{E}|X| < \infty.$$

On the event

$$\{|X| > x\},$$

we have

$$x < |X|.$$

Therefore,

$$x\mathbf{1}_{\{|X|>x\}} \leq |X|\mathbf{1}_{\{|X|>x\}}.$$

Taking expectations,

$$x\mathbb{P}(|X| > x) \leq \mathbb{E} \left[|X|\mathbf{1}_{\{|X|>x\}} \right].$$

For every ω ,

$$|X(\omega)|\mathbf{1}_{\{|X(\omega)|>x\}} \longrightarrow 0 \quad \text{as } x \rightarrow \infty.$$

Moreover,

$$0 \leq |X|\mathbf{1}_{\{|X|>x\}} \leq |X|,$$

and

$$\mathbb{E}|X| < \infty.$$

Therefore, by the dominated convergence theorem,

$$\mathbb{E} \left[|X|\mathbf{1}_{\{|X|>x\}} \right] \longrightarrow 0.$$

Consequently,

$$0 \leq x\mathbb{P}(|X| > x) \leq \mathbb{E} \left[|X|\mathbf{1}_{\{|X|>x\}} \right] \longrightarrow 0.$$

Hence

$$\boxed{x\mathbb{P}(|X| > x) \longrightarrow 0.}$$

The proof is complete. □

14.5.3 Relation Between Tail and Moment Conditions

Observation

The two preceding results give the implications

$$X \in L^1 \implies x\mathbb{P}(|X| > x) \longrightarrow 0,$$

while

$$x\mathbb{P}(|X| > x) \longrightarrow 0 \implies X \in L^p \text{ for every } 0 < p < 1.$$

Thus the tail condition lies between integrability and the collection of fractional moment conditions:

$$L^1 \implies \{x\mathbb{P}(|X| > x) \rightarrow 0\} \implies \bigcap_{0 < p < 1} L^p.$$

End of Lecture

Probability Theory

Lecture 15

Introduction to Probability

Date: Monday, 28 September 2026

Instructor: Dr. Nishant Chandgotia

15.1 Tails and the Behaviour of Sums

15.1.1 How Does S_n/n Behave for Large n ?

Let $X_1, X_2, \dots$ be a sequence of random variables and define

$$S_n = X_1 + \dots + X_n.$$

A fundamental question is:

Question 1. Behaviour of the Empirical Average

How does $\frac{S_n}{n}$ behave as $n \rightarrow \infty$?

The answer is strongly influenced by the tail behaviour of the random variables. In particular, the quantity $\mathbb{P}(|X| > t)$, $t \rightarrow \infty$, plays an important role.

Observation

The behaviour of the tails

$$\mathbb{P}(|X| > t)$$

for large t determines how frequently very large values of X occur.

Random variables whose tails decay slowly are often referred to as having *heavy tails*.

15.1.2 Normalization of Partial Sums

For suitable sequences of random variables, one may seek a sequence of positive numbers b_n such that $b_n \rightarrow \infty$ and

$$\frac{S_n}{b_n} \xrightarrow{P} 0.$$

The appropriate normalization depends on the tail behaviour of the random variables.

This motivates the study of weak laws for triangular arrays and appropriate choices of the normalization sequence b_n .

15.2 Fair Games and Pricing

15.2.1 Pricing an Independent Game

Suppose a casino game has winnings represented by a random variable X .

If a sequence of people independently plays the same game, the winnings give rise to an i.i.d. sequence

$$X_1, X_2, \dots, \quad X_i \stackrel{d}{=} X.$$

Let

$$S_n = X_1 + \dots + X_n.$$

We now ask:

Question 2. How Should the Casino Price the Game?

Suppose $\mathbb{E}(|X|) < \infty$.

Then the expected winning per game is $\mu = \mathbb{E}(X)$.

If the casino charges a price μ' per game, how should μ' be chosen?

15.2.2 Three Possible Pricing Scenarios

Suppose

$$\mu' > \mu.$$

Then, on average, the casino makes money.

However, the price is higher than the expected winning and may therefore be viewed as non-competitive.

If

$$\mu' < \mu,$$

then the casino loses money on average.

The natural fair price is therefore

$$\mu' = \mu = \mathbb{E}(X).$$

When

$$\mathbb{E}(|X|) < \infty,$$

the Weak Law of Large Numbers gives

$$\frac{S_n}{n} \xrightarrow{P} \mu.$$

Thus the total net loss of the casino after n games, when the price per game is μ , is

$$n\mu - S_n.$$

Observation

The Weak Law tells us that

$$\frac{n\mu - S_n}{n} = \mu - \frac{S_n}{n} \xrightarrow{P} 0.$$

Thus

$$n\mu - S_n = o_P(n).$$

The question is whether the casino's net loss $n\mu - S_n$ can nevertheless become large and positive.

15.2.3 What Happens When the Variance Is Finite?

Suppose additionally that

$$\text{Var}(X) = \sigma^2 < \infty.$$

Then the Central Limit Theorem gives

$$\frac{S_n - n\mu}{\sqrt{n}} \xrightarrow{d} N(0, \sigma^2).$$

Equivalently,

$$\frac{n\mu - S_n}{\sqrt{n}} \xrightarrow{d} N(0, \sigma^2).$$

Thus, in the finite-variance case, the fluctuations of the total winnings around their mean are typically of order

$$\sqrt{n}.$$

The limiting normal distribution is symmetric about zero, so the positive and negative fluctuations are balanced asymptotically in the sense described by the Central Limit Theorem.

Question 3. The Finite-Expectation Case

What happens if we only assume $X \in L^1$ and do not assume $X \in L^2$?

15.3 The St. Petersburg Game

15.3.1 Description of the Game

Consider the following game.

A fair coin is tossed repeatedly until the first head appears.

Let N denote the number of tosses required to obtain the first head.

The winnings are defined by

$$X = 2^N.$$

Since the first head occurs on the k -th toss exactly when the first $k - 1$ tosses are tails and the k -th toss is a head,

$$\mathbb{P}(N = k) = \left(\frac{1}{2}\right)^k.$$

Consequently,

$$\mathbb{P}(X = 2^k) = 2^{-k}, \quad k \geq 1.$$

15.3.2 The Expected Winning

We have

$$\begin{aligned} \mathbb{E}(X) &= \sum_{k=1}^{\infty} 2^k \mathbb{P}(X = 2^k) \\ &= \sum_{k=1}^{\infty} 2^k 2^{-k} \\ &= \sum_{k=1}^{\infty} 1 \\ &= \infty. \end{aligned}$$

Therefore,

$$\mathbb{E}(X) = \infty.$$

Thus the usual fair-price prescription price $= \mathbb{E}(X)$ does not give a finite price.

Observation

The St. Petersburg game illustrates the difficulty caused by a heavy-tailed distribution. Although extremely large winnings are rare, they are sufficiently large to make the expectation infinite.

15.3.3 Two Ways to Make the Game Finite

There are at least two natural ways to modify the game.

1. **Cap the winnings.**

Impose an upper bound on the amount that can be won.

2. **Limit the number of plays.**

Instead of considering infinitely many independent plays, consider only a finite number of plays and determine an appropriate price for them.

We now consider the second approach.

15.4 Pricing n Independent St. Petersburg Games

15.4.1 The Total Winning

Let

$$X_{n,1}, X_{n,2}, \dots, X_{n,n}$$

be independent copies of the St. Petersburg random variable X .

Thus

$$X_{n,k} \stackrel{d}{=} X, \quad 1 \leq k \leq n.$$

Define

$$S_n = \sum_{k=1}^n X_{n,k}.$$

We want to understand the correct scale of S_n .

Question 4. Pricing n Independent St. Petersburg Games

How should one price n independent plays of the St. Petersburg game?

15.4.2 Applying the Weak Law for Triangular Arrays

We use the Weak Law for triangular arrays.

We seek a sequence $b_n \rightarrow \infty$ such that

$$n\mathbb{P}(X > b_n) \rightarrow 0, \tag{1}$$

$$b_n^{-2} n \mathbb{E} \left[X^2 \mathbf{1}_{\{X \leq b_n\}} \right] \rightarrow 0. \tag{2}$$

These are precisely the two conditions needed to apply the triangular array weak law with $X_{n,k} = X_k$ and b_n as the normalization.

15.4.3 The Tail Probability

Let

$$L_n = \lfloor \log_2 b_n \rfloor.$$

Then

$$2^{L_n} \leq b_n < 2^{L_n+1}.$$

Therefore,

$$X > b_n$$

implies

$$X = 2^k$$

with

$$k > L_n.$$

Hence

$$\begin{aligned} \mathbb{P}(X > b_n) &= \sum_{k=L_n+1}^{\infty} 2^{-k} \\ &= 2^{-L_n}. \end{aligned}$$

Since

$$2^{L_n} \leq b_n < 2^{L_n+1},$$

we obtain

$$\boxed{\frac{1}{b_n} < \mathbb{P}(X > b_n) \leq \frac{2}{b_n}}. \quad (3)$$

15.4.4 Choice of b_n

Choose

$$\boxed{b_n = n2^{k(n)}},$$

where

$$k(n) \longrightarrow \infty.$$

Then, by (3),

$$\mathbb{P}(X > b_n) \leq \frac{2}{b_n}.$$

Therefore,

$$n\mathbb{P}(X > b_n) \leq \frac{2n}{b_n} = \frac{2}{2^{k(n)}} \longrightarrow 0.$$

Thus condition (1) is satisfied.

15.4.5 The Truncated Second Moment

Let

$$L_n = \lfloor \log_2 b_n \rfloor.$$

Then

$$\begin{aligned} \mathbb{E} \left[X^2 \mathbf{1}_{\{X \leq b_n\}} \right] &= \sum_{k=1}^{L_n} (2^k)^2 \mathbb{P}(X = 2^k) \\ &= \sum_{k=1}^{L_n} 2^{2k} 2^{-k} \\ &= \sum_{k=1}^{L_n} 2^k \\ &= 2^{L_n+1} - 2. \end{aligned}$$

Since

$$2^{L_n} \leq b_n,$$

we have

$$2^{L_n+1} \leq 2b_n.$$

Therefore,

$$\mathbb{E} \left[X^2 \mathbf{1}_{\{X \leq b_n\}} \right] \leq 2b_n.$$

Consequently,

$$\begin{aligned} b_n^{-2} n \mathbb{E} \left[X^2 \mathbf{1}_{\{X \leq b_n\}} \right] &\leq \frac{2n}{b_n} \\ &= \frac{2}{2^{k(n)}} \\ &\longrightarrow 0. \end{aligned}$$

Thus condition (2) is also satisfied.

15.4.6 Conclusion from the Weak Law

The Weak Law for triangular arrays therefore gives

$$\frac{S_n - a_n}{b_n} \xrightarrow{P} 0,$$

where

$$a_n = n\mathbb{E} \left[X \mathbf{1}_{\{X \leq b_n\}} \right].$$

Now

$$\begin{aligned} \mathbb{E} \left[X \mathbf{1}_{\{X \leq b_n\}} \right] &= \sum_{k=1}^{L_n} 2^k 2^{-k} \\ &= \sum_{k=1}^{L_n} 1 \\ &= L_n. \end{aligned}$$

Hence

$$a_n = nL_n = n \lfloor \log_2 b_n \rfloor.$$

Therefore,

$$\boxed{\frac{S_n - n \lfloor \log_2 b_n \rfloor}{b_n} \xrightarrow{P} 0.} \quad (4)$$

15.5 The Natural Scale of the St. Petersburg Game

15.5.1 Choosing b_n for $S_n/b_n \rightarrow 1$

Ideally, we would like to choose b_n such that

$$\boxed{\frac{S_n}{b_n} \xrightarrow{P} 1.}$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P} \left(\left| \frac{S_n}{b_n} - 1 \right| > \varepsilon \right) \rightarrow 0.$$

Using (4), it is enough to choose b_n such that

$$\frac{n \lfloor \log_2 b_n \rfloor}{b_n} \rightarrow 1.$$

Thus we require

$$\boxed{\frac{n \lfloor \log_2 b_n \rfloor}{b_n} \rightarrow 1.} \quad (5)$$

Now take

$$b_n = n2^{k(n)}.$$

Then

$$\frac{n \lfloor \log_2 b_n \rfloor}{b_n} = \frac{\lfloor \log_2 n + k(n) \rfloor}{2^{k(n)}}.$$

Therefore, we want

$$\frac{\log_2 n + k(n)}{2^{k(n)}} \rightarrow 1.$$

A natural choice is

$$k(n) = \log_2 \log_2 n.$$

Indeed,

$$2^{k(n)} = \log_2 n,$$

and therefore

$$\frac{\log_2 n + \log_2 \log_2 n}{\log_2 n} \rightarrow 1.$$

Thus we may choose

$$b_n = n \log_2 n.$$

Consequently,

$$\frac{S_n}{n \log_2 n} \xrightarrow{P} 1.$$

Observation

The natural scale of the total winnings in n independent St. Petersburg games is therefore

$$n \log_2 n.$$

Hence the natural price per game grows like

$$\log_2 n.$$

This is very different from the ordinary finite-expectation case, where the natural price is a fixed number $\mathbb{E}(X)$.

15.6 From Convergence in Probability to Almost Sure Convergence

15.6.1 A Bernoulli Example

Let

$$X_1, X_2, \dots$$

be i.i.d. Bernoulli random variables with

$$X_i \sim \text{Ber}\left(\frac{1}{2}\right).$$

Define

$$S_n = X_1 + \cdots + X_n.$$

The Weak Law gives

$$\frac{S_n}{n} \xrightarrow{P} \frac{1}{2}.$$

We now ask whether

$$\frac{S_n}{n} \longrightarrow \frac{1}{2}$$

almost surely.

Question 5. Almost Sure Convergence

Does

$$\frac{S_n}{n} \longrightarrow \frac{1}{2}$$

almost surely?

15.6.2 Exponential Probability Bounds

For every $\varepsilon > 0$, Hoeffding's inequality gives

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

In particular, there exists a constant $c > 0$ such that

$$\boxed{\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon\right) \leq 2e^{-cn}.} \quad (6)$$

15.6.3 Borel–Cantelli Argument

Define

$$A_n = \left\{\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon\right\}.$$

From (6),

$$\mathbb{P}(A_n) \leq 2e^{-cn}.$$

Therefore,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) \leq 2 \sum_{n=1}^{\infty} e^{-cn} < \infty.$$

Hence, by the first Borel–Cantelli lemma,

$$\mathbb{P}(A_n \text{ infinitely often}) = 0.$$

Equivalently,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon \text{ infinitely often}\right) = 0.$$

Thus, for every $\varepsilon > 0$,

$$\left|\frac{S_n}{n} - \frac{1}{2}\right| \leq \varepsilon$$

eventually almost surely.

Since this holds for every $\varepsilon > 0$,

$$\boxed{\frac{S_n}{n} \longrightarrow \frac{1}{2} \quad \text{almost surely.}}$$

Observation

This illustrates an important principle:

exponentially small deviation probabilities + Borel–Cantelli

$\implies$ almost sure convergence.

15.6.4 Equivalent Contradiction Argument

Suppose, for contradiction, that

$$\frac{S_n}{n} \not\rightarrow \frac{1}{2} \quad \text{almost surely.}$$

Then

$$\mathbb{P}\left(\frac{S_n}{n} \not\rightarrow \frac{1}{2}\right) > 0.$$

Equivalently, there exist

$$\theta > 0 \quad \text{and} \quad \varepsilon > 0$$

such that

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon \text{ infinitely often}\right) > \theta.$$

But the Borel–Cantelli argument above gives

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon \text{ infinitely often}\right) = 0,$$

which is a contradiction.

Therefore,

$$\boxed{\frac{S_n}{n} \longrightarrow \frac{1}{2} \quad \text{almost surely.}}$$

End of Lecture

Probability Theory

Lecture 16

Introduction to Probability

Date: Wednesday, 30 September 2026

Instructor: Dr. Nishant Chandgotia

16.1 Strong Law of Large Numbers

16.1.1 Strong Law for Bernoulli Random Variables

Let

$$X_1, X_2, \dots \stackrel{\text{i.i.d.}}{\sim} \text{Ber}\left(\frac{1}{2}\right),$$

and define

$$S_n = X_1 + \dots + X_n.$$

We want to prove that

$$\boxed{\frac{S_n}{n} \xrightarrow{\text{a.s.}} \frac{1}{2}.}$$

Fix $\varepsilon > 0$ and define the events

$$A_n = \left\{ \left| \frac{S_n}{n} - \frac{1}{2} \right| > \varepsilon \right\}.$$

For Bernoulli random variables, Hoeffding's inequality gives the exponential tail estimate

$$\mathbb{P}(A_n) = \mathbb{P}\left(\left| \frac{S_n}{n} - \frac{1}{2} \right| > \varepsilon\right) \leq C' e^{-Cn},$$

for some constants $C, C' > 0$ depending on ε .

Therefore,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) \leq C' \sum_{n=1}^{\infty} e^{-Cn} < \infty.$$

Hence, by the first Borel–Cantelli lemma,

$$\boxed{\mathbb{P}(A_n \text{ i.o.}) = 0.}$$

Equivalently, with probability one, only finitely many of the events A_n occur.

Thus, for almost every ω , there exists

$$N = N(\omega, \varepsilon)$$

such that

$$\left| \frac{S_n(\omega)}{n} - \frac{1}{2} \right| \leq \varepsilon \quad \text{for all } n \geq N.$$

Since $\varepsilon > 0$ was arbitrary,

$$\boxed{\frac{S_n}{n} \xrightarrow{\text{a.s.}} \frac{1}{2}.}$$

16.1.2 A Stronger Tail Estimate

The above argument can be written in a slightly stronger form.

For every n ,

$$\begin{aligned} & \mathbb{P} \left(\left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ for some } k > n \right) \\ &= \mathbb{P} \left(\bigcup_{k=n+1}^{\infty} A_k \right) \\ &\leq \sum_{k=n+1}^{\infty} \mathbb{P}(A_k), \end{aligned}$$

where the last inequality follows from the union bound.

Using

$$\mathbb{P}(A_k) \leq C' e^{-Ck},$$

we obtain

$$\begin{aligned} & \mathbb{P} \left(\left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ for some } k > n \right) \\ &\leq C' \sum_{k=n+1}^{\infty} e^{-Ck} \\ &= C' \frac{e^{-C(n+1)}}{1 - e^{-C}}. \end{aligned}$$

Define

$$C'' = \frac{C' e^{-C}}{1 - e^{-C}}.$$

Then

$$\boxed{\mathbb{P} \left(\left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ for some } k > n \right) \leq C'' e^{-Cn}.}$$

Since

$$C'' e^{-Cn} \rightarrow 0,$$

we obtain

$$\boxed{\mathbb{P} \left(\left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ for some } k > n \right) \rightarrow 0.}$$

Now observe that

$$\begin{aligned} & \left\{ \left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ infinitely often} \right\} \\ &= \bigcap_{n=1}^{\infty} \left\{ \left| \frac{S_k}{k} - \frac{1}{2} \right| > \varepsilon \text{ for some } k > n \right\}. \end{aligned}$$

The events on the right-hand side form a decreasing sequence. Therefore, by continuity of probability from above,

$$\begin{aligned} & \mathbb{P}\left(\left|\frac{S_k}{k} - \frac{1}{2}\right| > \varepsilon \text{ i.o.}\right) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}\left(\left|\frac{S_k}{k} - \frac{1}{2}\right| > \varepsilon \text{ for some } k > n\right) \\ &= 0. \end{aligned}$$

Thus,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| > \varepsilon \text{ i.o.}\right) = 0.$$

Since this holds for every $\varepsilon > 0$, we conclude that

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \frac{1}{2}.$$

16.2 Borel–Cantelli Lemmas

Let $(A_n)_{n \geq 1}$ be a sequence of events.

We say that A_n occurs *infinitely often*, abbreviated as “ A_n i.o.”, if infinitely many of the events A_n occur.

The event that A_n occurs infinitely often is denoted by

$$\{A_n \text{ i.o.}\}.$$

We have

$$\{A_n \text{ i.o.}\} = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n.$$

Indeed,

$$\omega \in \{A_n \text{ i.o.}\}$$

if and only if, for every $m \geq 1$, there exists some $n \geq m$ such that

$$\omega \in A_n.$$

Equivalently,

$$\{A_n \text{ i.o.}\} = \limsup_{n \rightarrow \infty} A_n.$$

Similarly,

$$\{A_n \text{ eventually}\} = \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} A_n.$$

That is, A_n occurs eventually if there exists some m such that A_n occurs for every $n \geq m$.

16.2.1 First Borel–Cantelli Lemma

Theorem 1. First Borel–Cantelli Lemma

Let $(A_n)_{n \geq 1}$ be a sequence of events. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\boxed{\mathbb{P}(A_n \text{ i.o.}) = 0.}$$

Proof. Recall that

$$\{A_n \text{ i.o.}\} = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n.$$

For each $m \geq 1$, define

$$B_m = \bigcup_{n=m}^{\infty} A_n.$$

Then

$$B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$$

and

$$\{A_n \text{ i.o.}\} = \bigcap_{m=1}^{\infty} B_m.$$

By the union bound,

$$\mathbb{P}(B_m) = \mathbb{P}\left(\bigcup_{n=m}^{\infty} A_n\right) \leq \sum_{n=m}^{\infty} \mathbb{P}(A_n).$$

Since

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

the tail of this convergent series tends to zero. Hence

$$\sum_{n=m}^{\infty} \mathbb{P}(A_n) \longrightarrow 0.$$

Therefore,

$$\mathbb{P}(B_m) \longrightarrow 0.$$

Since (B_m) is a decreasing sequence of events, continuity of probability from above gives

$$\begin{aligned} \mathbb{P}(A_n \text{ i.o.}) &= \mathbb{P}\left(\bigcap_{m=1}^{\infty} B_m\right) \\ &= \lim_{m \rightarrow \infty} \mathbb{P}(B_m) \\ &= 0. \end{aligned}$$

Thus

$$\boxed{\mathbb{P}(A_n \text{ i.o.}) = 0.}$$

□

16.2.2 Second Borel–Cantelli Lemma

Theorem 2. Second Borel–Cantelli Lemma

Let $(A_n)_{n \geq 1}$ be a sequence of independent events. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\boxed{\mathbb{P}(A_n \text{ i.o.}) = 1.}$$

Proof. We prove that the complementary event has probability zero.

The event that only finitely many of the A_n occur is

$$\{A_n \text{ only finitely often}\} = \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} A_n^c.$$

For each m , define

$$C_m = \bigcap_{n=m}^{\infty} A_n^c.$$

Then

$$C_1 \subseteq C_2 \subseteq C_3 \subseteq \cdots.$$

Hence, by continuity of probability from below,

$$\mathbb{P}\left(\bigcup_{m=1}^{\infty} C_m\right) = \lim_{m \rightarrow \infty} \mathbb{P}(C_m).$$

Now, for any integer $N \geq m$,

$$C_m \subseteq \bigcap_{n=m}^N A_n^c.$$

Therefore,

$$\mathbb{P}(C_m) \leq \mathbb{P}\left(\bigcap_{n=m}^N A_n^c\right).$$

Since the events A_n are independent, their complements are also independent. Thus

$$\begin{aligned} \mathbb{P}\left(\bigcap_{n=m}^N A_n^c\right) &= \prod_{n=m}^N \mathbb{P}(A_n^c) \\ &= \prod_{n=m}^N (1 - \mathbb{P}(A_n)). \end{aligned}$$

Using the elementary inequality

$$1 - x \leq e^{-x}, \quad 0 \leq x \leq 1,$$

we obtain

$$\begin{aligned}\mathbb{P}(C_m) &\leq \prod_{n=m}^N e^{-\mathbb{P}(A_n)} \\ &= \exp\left(-\sum_{n=m}^N \mathbb{P}(A_n)\right).\end{aligned}$$

Since

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

we have

$$\sum_{n=m}^N \mathbb{P}(A_n) \longrightarrow \infty \quad \text{as } N \rightarrow \infty.$$

Consequently,

$$\exp\left(-\sum_{n=m}^N \mathbb{P}(A_n)\right) \longrightarrow 0.$$

Hence

$$\mathbb{P}(C_m) = 0.$$

Therefore,

$$\begin{aligned}\mathbb{P}(A_n \text{ only finitely often}) &= \mathbb{P}\left(\bigcup_{m=1}^{\infty} C_m\right) \\ &= \lim_{m \rightarrow \infty} \mathbb{P}(C_m) \\ &= 0.\end{aligned}$$

Since

$$\{A_n \text{ i.o.}\}^c = \{A_n \text{ only finitely often}\},$$

we obtain

$$\mathbb{P}(A_n \text{ i.o.}) = 1 - \mathbb{P}(A_n \text{ only finitely often}) = 1.$$

Thus

$$\boxed{\mathbb{P}(A_n \text{ i.o.}) = 1.}$$

□

16.2.3 Comparison of the Two Borel–Cantelli Lemmas

The two Borel–Cantelli lemmas can be summarized as follows.

Theorem 3. Borel–Cantelli Summary

For a sequence of events $(A_n)_{n \geq 1}$,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 0.$$

If, in addition, the events are independent, then

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 1.$$

Observation

Independence is *not* required for the first Borel–Cantelli lemma, whereas independence is required for the second Borel–Cantelli lemma.

16.3 Strong Law Under a Fourth-Moment Assumption

16.3.1 Statement of the Result

Theorem 4. Strong Law with Finite Fourth Moment

Let $X_1, X_2, \dots$ be independent and identically distributed random variables satisfying $\mathbb{E}[X_1] = \mu$ and

$$\mathbb{E}[X_1^4] < \infty.$$

If

$$S_n = X_1 + \dots + X_n,$$

then

$$\boxed{\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu.}$$

16.3.2 Proof

Proof. By replacing X_i by $X_i - \mu$, it is enough to prove the result when

$$\mu = 0.$$

Thus assume

$$\mathbb{E}[X_i] = 0$$

and

$$\mathbb{E}[X_i^4] < \infty.$$

We shall prove that

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0.$$

Consider the fourth moment of S_n :

$$\mathbb{E}(S_n^4) = \mathbb{E} \left[\left(\sum_{i=1}^n X_i \right)^4 \right].$$

Expanding the fourth power gives

$$S_n^4 = \sum_{i_1, i_2, i_3, i_4=1}^n X_{i_1} X_{i_2} X_{i_3} X_{i_4}.$$

Because the X_i are independent and

$$\mathbb{E}[X_i] = 0,$$

any term containing an index that appears only once has expectation zero.

The only nonzero contributions are therefore of the following two types:

1. all four indices are equal;
2. two distinct indices each occur twice.

For the first type,

$$\sum_{i=1}^n \mathbb{E}[X_i^4] = n\mathbb{E}[X_1^4].$$

For the second type, choose two distinct indices $i \neq j$. The corresponding term is

$$X_i^2 X_j^2.$$

For each pair $\{i, j\}$ there are

$$\frac{4!}{2!2!} = 6$$

ways to arrange the indices. Hence the contribution is

$$6 \sum_{1 \leq i < j \leq n} \mathbb{E}[X_i^2 X_j^2].$$

By independence,

$$\mathbb{E}[X_i^2 X_j^2] = \mathbb{E}[X_i^2] \mathbb{E}[X_j^2].$$

Since the variables are identically distributed,

$$\mathbb{E}[X_i^2 X_j^2] = \left(\mathbb{E}[X_1^2] \right)^2.$$

Therefore,

$$\begin{aligned} \mathbb{E}[S_n^4] &= n\mathbb{E}[X_1^4] + 6 \binom{n}{2} \left(\mathbb{E}[X_1^2] \right)^2 \\ &= n\mathbb{E}[X_1^4] + 3n(n-1) \left(\mathbb{E}[X_1^2] \right)^2. \end{aligned}$$

Hence there exists a constant $C < \infty$ such that

$$\boxed{\mathbb{E}[S_n^4] \leq Cn^2.}$$

Now apply Markov's inequality to the nonnegative random variable S_n^4 . For every $\varepsilon > 0$,

$$\begin{aligned} \mathbb{P} \left(\left| \frac{S_n}{n} \right| > \varepsilon \right) &= \mathbb{P}(|S_n| > \varepsilon n) \\ &= \mathbb{P}(S_n^4 > \varepsilon^4 n^4) \\ &\leq \frac{\mathbb{E}[S_n^4]}{\varepsilon^4 n^4}. \end{aligned}$$

Using

$$\mathbb{E}[S_n^4] \leq Cn^2,$$

we obtain

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) \leq \frac{C}{\varepsilon^4 n^2}.$$

Therefore,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) \leq \frac{C}{\varepsilon^4} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

By the first Borel–Cantelli lemma,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon \text{ i.o.}\right) = 0.$$

Thus, for every $\varepsilon > 0$,

$$\left|\frac{S_n}{n}\right| \leq \varepsilon \quad \text{eventually almost surely.}$$

Hence

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0.$$

Returning to the original variables, write

$$X_i = (X_i - \mu) + \mu.$$

Then

$$\frac{S_n}{n} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu) + \mu.$$

The first term converges almost surely to 0. Therefore,

$$\boxed{\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu.}$$

□

End of Lecture

Quiz 7

Let $X_1, X_2, \dots$ be i.i.d. real-valued random variables and let $S_n = X_1 + \dots + X_n$. Suppose that there is a sequence of constants $a_n \in \mathbb{R}$ such that

$$\frac{S_n}{n} - a_n \xrightarrow{P} 0.$$

Prove that

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0 \quad \text{as } x \rightarrow \infty.$$

(a) Let $X'_1, X'_2, \dots$ be an independent copy of the sequence $(X_i)_{i \geq 1}$ and set

$$Y_i = X_i - X'_i.$$

Show that the Y_i 's are i.i.d. and symmetric, and that

$$\frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{P} 0.$$

(b) **A maximal inequality.**

Let

$$T_k = Y_1 + \dots + Y_k, \quad 1 \leq k \leq n.$$

Fix $t > 0$.

(i) Define the first crossing time

$$\tau = \min\{k : T_k > t\},$$

with $\tau = \infty$ if the set is empty.

On the event $\{\tau \leq n\}$, consider the transformation which reflects all the increments after the first crossing:

$$(Y_1, \dots, Y_\tau, Y_{\tau+1}, \dots, Y_n) \longmapsto (Y_1, \dots, Y_\tau, -Y_{\tau+1}, \dots, -Y_n).$$

Explain why this transformation preserves probabilities.

(ii) If $\tilde{T}_n$ denotes the endpoint after reflection, show that

$$\tilde{T}_n = 2T_\tau - T_n.$$

Deduce that

$$\mathbb{P}\left(\max_{1 \leq k \leq n} T_k > t\right) \leq 2\mathbb{P}(T_n > t).$$

(iii) Apply the same argument to $-T_k$ to show that

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |T_k| > t\right) \leq 2\mathbb{P}(|T_n| > t).$$

(iv) Show that

$$\max_{1 \leq i \leq n} |Y_i| > 2t \implies \max_{1 \leq k \leq n} |T_k| > t.$$

Conclude that

$$\mathbb{P}\left(\max_{1 \leq i \leq n} |Y_i| > 2t\right) \leq 2\mathbb{P}\left(\left|\sum_{i=1}^n Y_i\right| > t\right).$$

(c) Put

$$p_n = \mathbb{P}(|Y_1| > \varepsilon n).$$

Show that

$$\mathbb{P}\left(\max_{1 \leq i \leq n} |Y_i| > \varepsilon n\right) = 1 - (1 - p_n)^n.$$

Deduce that

$$n\mathbb{P}(|X_1 - X'_1| > \varepsilon n) \longrightarrow 0.$$

(d) Show that there exists $M < \infty$ such that

$$\mathbb{P}(|X'_1| \leq M) \geq \frac{1}{2}.$$

Show that, for $x > 2M$,

$$\{|X_1| > x, |X'_1| \leq M\} \subseteq \{|X_1 - X'_1| > x/2\}.$$

Hence prove that

$$\leq 2\mathbb{P}\left(|X_1 - X'_1| > \frac{x}{2}\right), \quad x > 2M.$$

(e) Combine the preceding parts to conclude that

$$x\mathbb{P}(|X_1| > x) \longrightarrow 0 \quad \text{as } x \rightarrow \infty.$$

Probability Theory

Lecture 17

Introduction to Probability

Date: Wednesday, 7 October 2026

Instructor: Dr. Nishant Chandgotia

17.1 Subsequence Characterization of Convergence in Probability

Theorem 1. Subsequence Characterization of Convergence in Probability

Let $X, X_1, X_2, \dots$ be random variables. Then

$$X_n \xrightarrow{P} X$$

if and only if every subsequence (X_{n_k}) has a further subsequence $(X_{n_{k_j}})$ such that

$$X_{n_{k_j}} \xrightarrow{\text{a.s.}} X.$$

Proof. We prove the two implications separately.

($\Rightarrow$) Suppose that $X_n \xrightarrow{P} X$.

We have to prove that every subsequence of (X_n) has a further subsequence which converges to X almost surely.

Let

$$(X_{n_k})$$

be an arbitrary subsequence of (X_n) .

Since

$$X_n \xrightarrow{P} X,$$

by definition, for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0.$$

We shall construct a further subsequence

$$X_{n_{k_1}}, X_{n_{k_2}}, \dots$$

such that

$$\mathbb{P}\left(|X_{n_{k_j}} - X| > \frac{1}{2^j}\right) < \frac{1}{2^j}$$

for every $j \geq 1$.

To see that this is possible, first choose k_1 such that

$$\mathbb{P}\left(|X_{n_{k_1}} - X| > 1\right) < \frac{1}{2}.$$

This is possible because

$$\mathbb{P}(|X_n - X| > 1) \longrightarrow 0.$$

Having chosen $k_1, \dots, k_{j-1}$, choose $k_j > k_{j-1}$ such that

$$\mathbb{P}\left(|X_{n_{k_j}} - X| > \frac{1}{2^j}\right) < \frac{1}{2^j}.$$

Again, this is possible because

$$\mathbb{P}\left(|X_n - X| > \frac{1}{2^j}\right) \longrightarrow 0.$$

Thus we have constructed a further subsequence satisfying

$$\mathbb{P}\left(|X_{n_{k_j}} - X| > \frac{1}{2^j}\right) < \frac{1}{2^j}.$$

Define the events

$$A_j = \left\{|X_{n_{k_j}} - X| > \frac{1}{2^j}\right\}.$$

Then

$$\sum_{j=1}^{\infty} \mathbb{P}(A_j) \leq \sum_{j=1}^{\infty} \frac{1}{2^j} = 1 < \infty.$$

Therefore, by the first Borel–Cantelli lemma,

$$\mathbb{P}(A_j \text{ i.o.}) = 0.$$

Hence, with probability one, only finitely many of the events A_j occur.

Thus, for almost every ω , there exists $J = J(\omega)$ such that for all $j \geq J$,

$$|X_{n_{k_j}}(\omega) - X(\omega)| \leq \frac{1}{2^j}.$$

Since

$$\frac{1}{2^j} \longrightarrow 0,$$

we obtain

$$X_{n_{k_j}}(\omega) \longrightarrow X(\omega)$$

for almost every ω .

Therefore,

$$\boxed{X_{n_{k_j}} \xrightarrow{\text{a.s.}} X.}$$

Since the original subsequence (X_{n_k}) was arbitrary, every subsequence has a further subsequence which converges almost surely to X .

($\Leftarrow$) Suppose that every subsequence of (X_n) has a further subsequence which converges almost surely to X .

We prove that

$$X_n \xrightarrow{P} X.$$

Suppose, to the contrary, that

$$X_n \not\xrightarrow{P} X.$$

By the definition of convergence in probability, this means that there exists some $\varepsilon_0 > 0$ for which

$$\mathbb{P}(|X_n - X| > \varepsilon_0)$$

does not converge to 0.

Therefore, there exists some $\delta > 0$ and a subsequence

$$X_{n_k}$$

such that

$$\mathbb{P}(|X_{n_k} - X| > \varepsilon_0) \geq \delta$$

for every $k \geq 1$.

Indeed, if no such δ and subsequence existed, then

$$\mathbb{P}(|X_n - X| > \varepsilon_0) \longrightarrow 0,$$

which would contradict our assumption.

By the hypothesis of the theorem, this subsequence (X_{n_k}) has a further subsequence

$$X_{n_{k_j}}$$

such that

$$X_{n_{k_j}} \xrightarrow{\text{a.s.}} X.$$

We now show that almost sure convergence implies convergence in probability.

Since

$$X_{n_{k_j}} \xrightarrow{\text{a.s.}} X,$$

for almost every ω ,

$$|X_{n_{k_j}}(\omega) - X(\omega)| \longrightarrow 0.$$

Fix $\varepsilon_0 > 0$ and define

$$B_m = \bigcup_{j=m}^{\infty} \{|X_{n_{k_j}} - X| > \varepsilon_0\}.$$

The events satisfy

$$B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots.$$

Moreover, since

$$X_{n_{k_j}} \xrightarrow{\text{a.s.}} X,$$

only finitely many of the events

$$\{|X_{n_{k_j}} - X| > \varepsilon_0\}$$

occur almost surely. Hence

$$\mathbb{P}\left(\bigcap_{m=1}^{\infty} B_m\right) = 0.$$

By continuity of probability from above,

$$\mathbb{P}(B_m) \longrightarrow \mathbb{P}\left(\bigcap_{m=1}^{\infty} B_m\right) = 0.$$

But for every $j \geq m$,

$$\{|X_{n_{k_j}} - X| > \varepsilon_0\} \subseteq B_m.$$

Therefore,

$$\mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) \leq \mathbb{P}(B_m).$$

For fixed m , this holds for every $j \geq m$. Hence

$$\limsup_{j \rightarrow \infty} \mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) \leq \mathbb{P}(B_m).$$

Letting $m \rightarrow \infty$, we get

$$\limsup_{j \rightarrow \infty} \mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) = 0.$$

Thus

$$\mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) \longrightarrow 0.$$

However, $(X_{n_{k_j}})$ is a further subsequence of (X_{n_k}) , and we constructed (X_{n_k}) so that

$$\mathbb{P}(|X_{n_k} - X| > \varepsilon_0) \geq \delta$$

for every k .

Consequently,

$$\mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) \geq \delta$$

for every j , which contradicts

$$\mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon_0) \longrightarrow 0.$$

This contradiction shows that our assumption

$$X_n \xrightarrow{P} X$$

was false. Therefore,

$$\boxed{X_n \xrightarrow{P} X.}$$

Hence,

$$\boxed{X_n \xrightarrow{P} X \iff \text{every subsequence of } (X_n) \text{ has a further subsequence converging a.s. to } X.}$$

□

17.2 Infinite First Moment

Theorem 2. Durrett's Theorem 2.3.8

Let $X_1, X_2, \dots$ be independent and identically distributed random variables such that

$$\mathbb{E}|X_i| = \infty.$$

Then

$$\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1.$$

Consequently, if

$$S_n = X_1 + \dots + X_n,$$

then

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} \text{ exists in } (-\infty, \infty)\right) = 0.$$

Proof. We divide the proof into two parts.

Part I: Proving that $|X_n| \geq n$ infinitely often almost surely.

For any nonnegative random variable Y , the tail-integral formula gives

$$\mathbb{E}Y = \int_0^\infty \mathbb{P}(Y > x) dx.$$

Applying this with $Y = |X_1|$, we obtain

$$\mathbb{E}|X_1| = \int_0^\infty \mathbb{P}(|X_1| > x) dx.$$

Now divide the integral into intervals $[n, n+1)$:

$$\mathbb{E}|X_1| = \sum_{n=0}^\infty \int_n^{n+1} \mathbb{P}(|X_1| > x) dx.$$

Since the function

$$x \mapsto \mathbb{P}(|X_1| > x)$$

is nonincreasing, for $x \in [n, n+1)$ we have

$$\mathbb{P}(|X_1| > x) \leq \mathbb{P}(|X_1| > n).$$

Therefore,

$$\begin{aligned} \mathbb{E}|X_1| &\leq \sum_{n=0}^\infty \int_n^{n+1} \mathbb{P}(|X_1| > n) dx \\ &= \sum_{n=0}^\infty \mathbb{P}(|X_1| > n). \end{aligned}$$

But by assumption,

$$\mathbb{E}|X_1| = \infty.$$

Hence

$$\sum_{n=0}^\infty \mathbb{P}(|X_1| > n) = \infty.$$

Since the random variables $X_1, X_2, \dots$ are identically distributed,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) = \infty.$$

Define the events

$$A_n = \{|X_n| > n\}.$$

Because $X_1, X_2, \dots$ are independent, the events $A_1, A_2, \dots$ are independent.

Therefore, by the second Borel–Cantelli lemma,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 1.$$

Consequently,

$$\mathbb{P}(|X_n| > n \text{ i.o.}) = 1.$$

Since

$$\{|X_n| > n \text{ i.o.}\} \subseteq \{|X_n| \geq n \text{ i.o.}\},$$

we obtain

$$\boxed{\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1.}$$

Part II: Proving that S_n/n cannot converge to a finite limit.

Let

$$C = \left\{ \omega : \lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} \text{ exists in } (-\infty, \infty) \right\}.$$

We will prove that

$$\mathbb{P}(C) = 0.$$

Suppose, for contradiction, that

$$\omega \in C.$$

Then, by definition, there exists some finite real number $L = L(\omega)$ such that

$$\frac{S_n(\omega)}{n} \longrightarrow L.$$

In particular, the sequence $S_n(\omega)/n$ is bounded. Hence there exists a finite constant $M = M(\omega)$ such that

$$\left| \frac{S_n(\omega)}{n} \right| \leq M$$

for all sufficiently large n .

We now consider the difference between two consecutive terms:

$$\frac{S_n}{n} - \frac{S_{n+1}}{n+1}.$$

Since

$$S_{n+1} = S_n + X_{n+1},$$

we have

$$\begin{aligned}
 \frac{S_n}{n} - \frac{S_{n+1}}{n+1} &= \frac{S_n}{n} - \frac{S_n + X_{n+1}}{n+1} \\
 &= \frac{(n+1)S_n - n(S_n + X_{n+1})}{n(n+1)} \\
 &= \frac{S_n - nX_{n+1}}{n(n+1)} \\
 &= \frac{S_n}{n(n+1)} - \frac{X_{n+1}}{n+1}.
 \end{aligned}$$

Thus

$$\boxed{\frac{S_n}{n} - \frac{S_{n+1}}{n+1} = \frac{S_n}{n(n+1)} - \frac{X_{n+1}}{n+1}.}$$

Because $\omega \in C$,

$$\frac{S_n(\omega)}{n} \longrightarrow L.$$

Therefore,

$$\frac{S_n(\omega)}{n(n+1)} = \frac{1}{n+1} \frac{S_n(\omega)}{n} \longrightarrow 0.$$

Hence

$$\boxed{\frac{S_n(\omega)}{n(n+1)} \longrightarrow 0.}$$

Now define

$$A = \{\omega : |X_n(\omega)| \geq n \text{ infinitely often}\}.$$

From Part I,

$$\mathbb{P}(A) = 1.$$

Suppose now that

$$\omega \in A \cap C.$$

Since $\omega \in A$,

$$|X_n(\omega)| \geq n$$

for infinitely many n .

Replacing n by $n+1$, this means

$$|X_{n+1}(\omega)| \geq n+1$$

for infinitely many n .

Consequently, for infinitely many n ,

$$\frac{|X_{n+1}(\omega)|}{n+1} \geq 1.$$

Using the identity above and the reverse triangle inequality,

$$\begin{aligned}
 \left| \frac{S_n(\omega)}{n} - \frac{S_{n+1}(\omega)}{n+1} \right| &= \left| \frac{S_n(\omega)}{n(n+1)} - \frac{X_{n+1}(\omega)}{n+1} \right| \\
 &\geq \frac{|X_{n+1}(\omega)|}{n+1} - \frac{|S_n(\omega)|}{n(n+1)}.
 \end{aligned}$$

For infinitely many n ,

$$\frac{|X_{n+1}(\omega)|}{n+1} \geq 1.$$

Moreover,

$$\frac{|S_n(\omega)|}{n(n+1)} \rightarrow 0.$$

Therefore, for all sufficiently large n among those infinitely many indices,

$$\frac{|S_n(\omega)|}{n(n+1)} < \frac{1}{3}.$$

Hence, infinitely often,

$$\left| \frac{S_n(\omega)}{n} - \frac{S_{n+1}(\omega)}{n+1} \right| \geq 1 - \frac{1}{3} = \frac{2}{3}.$$

Thus

$$\boxed{\left| \frac{S_n(\omega)}{n} - \frac{S_{n+1}(\omega)}{n+1} \right| > \frac{2}{3} \quad \text{i.o.}}$$

However, this contradicts $\omega \in C$.

Indeed, if

$$\frac{S_n(\omega)}{n} \rightarrow L,$$

then also

$$\frac{S_{n+1}(\omega)}{n+1} \rightarrow L.$$

Therefore,

$$\left| \frac{S_n(\omega)}{n} - \frac{S_{n+1}(\omega)}{n+1} \right| \leq \left| \frac{S_n(\omega)}{n} - L \right| + \left| \frac{S_{n+1}(\omega)}{n+1} - L \right| \rightarrow 0.$$

Thus the difference must converge to 0, which is impossible if it is larger than $2/3$ infinitely often.

We have therefore proved that

$$A \cap C = \emptyset.$$

Since

$$\mathbb{P}(A) = 1,$$

we have

$$\mathbb{P}(A^c) = 0.$$

Because

$$C \subseteq A^c,$$

it follows that

$$\mathbb{P}(C) \leq \mathbb{P}(A^c) = 0.$$

Hence

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} \text{ exists in } (-\infty, \infty)\right) = 0.$$

This completes the proof. □

17.3 Strong Law for Pairwise Independent Events

Theorem 3. Strong Law for Pairwise Independent Events

If $A_1, A_2, \dots$ are pairwise independent and

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\frac{\sum_{m=1}^n \mathbf{1}_{A_m}}{\sum_{m=1}^n \mathbb{P}(A_m)} \xrightarrow{\text{a.s.}} 1.$$

Proof. Let

$$X_m = \mathbf{1}_{A_m},$$

and define

$$S_n = X_1 + \dots + X_n = \sum_{m=1}^n \mathbf{1}_{A_m}.$$

Since the events A_m are pairwise independent, the random variables $X_m = \mathbf{1}_{A_m}$ are pairwise independent. Hence they are uncorrelated, and therefore

$$\text{Var}(S_n) = \sum_{m=1}^n \text{Var}(X_m).$$

Now $X_m \in \{0, 1\}$, so

$$X_m^2 = X_m.$$

Therefore,

$$\mathbb{E}(X_m^2) = \mathbb{E}(X_m)$$

and hence

$$\begin{aligned} \text{Var}(X_m) &= \mathbb{E}(X_m^2) - (\mathbb{E}X_m)^2 \\ &= \mathbb{E}(X_m) - (\mathbb{E}X_m)^2 \\ &\leq \mathbb{E}(X_m). \end{aligned}$$

Thus

$$\begin{aligned}\mathrm{Var}(S_n) &= \sum_{m=1}^n \mathrm{Var}(X_m) \\ &\leq \sum_{m=1}^n \mathbb{E}(X_m) \\ &= \mathbb{E}(S_n).\end{aligned}$$

Since

$$\mathbb{E}(X_m) = \mathbb{P}(A_m),$$

we have

$$\mathbb{E}(S_n) = \sum_{m=1}^n \mathbb{P}(A_m).$$

By assumption,

$$\sum_{m=1}^{\infty} \mathbb{P}(A_m) = \infty,$$

and hence

$$\mathbb{E}(S_n) \longrightarrow \infty.$$

We now use Chebyshev's inequality. For every $\delta > 0$,

$$\begin{aligned}\mathbb{P}(|S_n - \mathbb{E}S_n| > \delta \mathbb{E}S_n) &\leq \frac{\mathrm{Var}(S_n)}{\delta^2 (\mathbb{E}S_n)^2} \\ &\leq \frac{\mathbb{E}S_n}{\delta^2 (\mathbb{E}S_n)^2} \\ &= \frac{1}{\delta^2 \mathbb{E}S_n}.\end{aligned}$$

Thus

$$\mathbb{P}(|S_n - \mathbb{E}S_n| > \delta \mathbb{E}S_n) \leq \frac{1}{\delta^2 \mathbb{E}S_n} \longrightarrow 0.$$

Therefore,

$$\frac{S_n}{\mathbb{E}S_n} \xrightarrow{P} 1.$$

That is, we have convergence in probability.

To obtain almost sure convergence, we use a suitable subsequence.

Construction of the subsequence.

Since

$$\mathbb{E}S_n \longrightarrow \infty,$$

for each $k \geq 1$ define

$$n_k = \inf \left\{ n : \mathbb{E}S_n \geq k^2 \right\}.$$

This is well-defined because $\mathbb{E}S_n \rightarrow \infty$.

By the definition of n_k ,

$$\mathbb{E}S_{n_k} \geq k^2.$$

Furthermore, since n_k is the first index for which $\mathbb{E}S_n \geq k^2$, we have

$$\mathbb{E}S_{n_k-1} < k^2.$$

Since

$$\mathbb{E}X_m = \mathbb{P}(A_m) \leq 1,$$

we have

$$\mathbb{E}S_{n_k} = \mathbb{E}S_{n_k-1} + \mathbb{E}X_{n_k} < k^2 + 1.$$

Consequently,

$$\boxed{k^2 \leq \mathbb{E}S_{n_k} < k^2 + 1.}$$

For convenience, define

$$T_k = S_{n_k}.$$

Applying the previous Chebyshev estimate with $n = n_k$ gives

$$\mathbb{P}(|T_k - \mathbb{E}T_k| > \delta \mathbb{E}T_k) \leq \frac{1}{\delta^2 \mathbb{E}T_k}.$$

Since

$$\mathbb{E}T_k = \mathbb{E}S_{n_k} \geq k^2,$$

we obtain

$$\boxed{\mathbb{P}(|T_k - \mathbb{E}T_k| > \delta \mathbb{E}T_k) \leq \frac{1}{\delta^2 k^2}.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(|T_k - \mathbb{E}T_k| > \delta \mathbb{E}T_k) &\leq \frac{1}{\delta^2} \sum_{k=1}^{\infty} \frac{1}{k^2} \\ &< \infty. \end{aligned}$$

By the first Borel–Cantelli lemma,

$$\mathbb{P}(|T_k - \mathbb{E}T_k| > \delta \mathbb{E}T_k \text{ i.o.}) = 0.$$

Thus, almost surely,

$$|T_k - \mathbb{E}T_k| \leq \delta \mathbb{E}T_k$$

for all sufficiently large k .

Equivalently,

$$1 - \delta \leq \frac{T_k}{\mathbb{E}T_k} \leq 1 + \delta$$

eventually almost surely.

Since $\delta > 0$ is arbitrary, we conclude that

$$\boxed{\frac{T_k}{\mathbb{E}T_k} \xrightarrow{\text{a.s.}} 1.}$$

Interpolation between the subsequence points.

It remains to prove that

$$\frac{S_n}{\mathbb{E}S_n} \xrightarrow{\text{a.s.}} 1$$

for the full sequence n , not only for the subsequence n_k .

Since

$$\mathbb{E}S_n = \sum_{m=1}^n \mathbb{P}(A_m)$$

is nondecreasing in n , and S_n is also nondecreasing because $X_m = \mathbf{1}_{A_m} \geq 0$, we have, whenever

$$n_k \leq n < n_{k+1},$$

that

$$T_k = S_{n_k} \leq S_n \leq S_{n_{k+1}} = T_{k+1}.$$

Also,

$$\mathbb{E}T_k = \mathbb{E}S_{n_k} \leq \mathbb{E}S_n \leq \mathbb{E}S_{n_{k+1}} = \mathbb{E}T_{k+1}.$$

Therefore,

$$\frac{T_k}{\mathbb{E}T_{k+1}} \leq \frac{S_n}{\mathbb{E}S_n} \leq \frac{T_{k+1}}{\mathbb{E}T_k}.$$

We rewrite the two bounds as

$$\begin{aligned} \frac{T_k}{\mathbb{E}T_{k+1}} &= \frac{\mathbb{E}T_k}{\mathbb{E}T_{k+1}} \frac{T_k}{\mathbb{E}T_k}, \\ \frac{T_{k+1}}{\mathbb{E}T_k} &= \frac{T_{k+1}}{\mathbb{E}T_{k+1}} \frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k}. \end{aligned}$$

Thus

$$\boxed{\frac{\mathbb{E}T_k}{\mathbb{E}T_{k+1}} \frac{T_k}{\mathbb{E}T_k} \leq \frac{S_n}{\mathbb{E}S_n} \leq \frac{T_{k+1}}{\mathbb{E}T_{k+1}} \frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k}}.$$

We already know that

$$\frac{T_k}{\mathbb{E}T_k} \xrightarrow{\text{a.s.}} 1$$

and

$$\frac{T_{k+1}}{\mathbb{E}T_{k+1}} \xrightarrow{\text{a.s.}} 1.$$

It therefore remains only to show that

$$\frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k} \longrightarrow 1.$$

From the construction of n_k ,

$$k^2 \leq \mathbb{E}T_k < k^2 + 1$$

and

$$(k+1)^2 \leq \mathbb{E}T_{k+1} < (k+1)^2 + 1.$$

Hence

$$\frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k} \leq \frac{(k+1)^2 + 1}{k^2}.$$

But

$$\frac{(k+1)^2 + 1}{k^2} = \frac{k^2 + 2k + 2}{k^2} = 1 + \frac{2}{k} + \frac{2}{k^2} \longrightarrow 1.$$

Also,

$$\frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k} \geq 1,$$

because $\mathbb{E}T_k$ is nondecreasing.

Therefore,

$$\boxed{\frac{\mathbb{E}T_{k+1}}{\mathbb{E}T_k} \longrightarrow 1.}$$

Consequently,

$$\frac{\mathbb{E}T_k}{\mathbb{E}T_{k+1}} \longrightarrow 1.$$

Hence, almost surely,

$$\frac{T_k}{\mathbb{E}T_{k+1}} \longrightarrow 1$$

and

$$\frac{T_{k+1}}{\mathbb{E}T_k} \longrightarrow 1.$$

Therefore, by the squeeze theorem,

$$\boxed{\frac{S_n}{\mathbb{E}S_n} \xrightarrow{\text{a.s.}} 1.}$$

Finally, since

$$S_n = \sum_{m=1}^n \mathbf{1}_{A_m}$$

and

$$\mathbb{E}S_n = \sum_{m=1}^n \mathbb{P}(A_m),$$

we obtain

$$\boxed{\frac{\sum_{m=1}^n \mathbf{1}_{A_m}}{\sum_{m=1}^n \mathbb{P}(A_m)} \xrightarrow{\text{a.s.}} 1.}$$

□

End of Lecture

Feedback and Corrections

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Your feedback is greatly appreciated and will help improve future versions of these notes.

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